



Deep Reinforcement Learning

HOMEWORK 3 — ACTOR-CRITIC METHODS

1 Deterministic Actor-Critic: From DPG to TD3

1.1 Background: Stochastic Policy Gradients

Stochastic Policy Gradient Theorem

For a stochastic policy $\pi_\theta(a|s)$, the gradient of the performance objective

$$J(\theta) = \mathbb{E}_{s \sim \rho^{\pi_\theta}} [v^{\pi_\theta}(s)]$$

is

$$\nabla_\theta J(\theta) = \mathbb{E}_{s \sim \rho^{\pi_\theta}, a \sim \pi_\theta} [\nabla_\theta \log \pi_\theta(a|s) Q^{\pi_\theta}(s, a)].$$

In stochastic actor-critic methods, the critic parameters w approximate the action-value function $Q^w(s, a)$ while the actor parameters θ optimize the policy $\pi_\theta(a|s)$ using temporal-difference (TD) errors. The parameter update configurations depend on the data collection mechanism:

1. On-Policy Stochastic Actor-Critic

Transitions are sampled sequentially from the current target policy, meaning $a_t \sim \pi_\theta(\cdot|s_t)$ and $a_{t+1} \sim \pi_\theta(\cdot|s_{t+1})$. The parameters are updated as follows:

- ◇ **TD Error:** $\delta_t = r_t + \gamma Q^w(s_{t+1}, a_{t+1}) - Q^w(s_t, a_t)$
- ◇ **Critic Update:** $w_{t+1} = w_t + \alpha_w \delta_t \nabla_w Q^w(s_t, a_t)$
- ◇ **Actor Update:** $\theta_{t+1} = \theta_t + \alpha_\theta \nabla_\theta \log \pi_\theta(a_t|s_t) Q^w(s_t, a_t)$

2. Off-Policy Stochastic Actor-Critic

Transitions are generated by a separate behavior policy $\beta(a|s)$, where $a_t \sim \beta(\cdot|s_t)$ and $a_{t+1} \sim \beta(\cdot|s_{t+1})$. To correct for the distribution mismatch, an importance sampling ratio $\rho_t = \frac{\pi_\theta(a_t|s_t)}{\beta(a_t|s_t)}$ is incorporated into the policy update :

- ◇ **TD Error:** $\delta_t = r_t + \gamma Q^w(s_{t+1}, a_{t+1}) - Q^w(s_t, a_t)$
- ◇ **Critic Update:** $w_{t+1} = w_t + \alpha_w \delta_t \nabla_w Q^w(s_t, a_t)$
- ◇ **Actor Update:** $\theta_{t+1} = \theta_t + \alpha_\theta \rho_t \nabla_\theta \log \pi_\theta(a_t|s_t) Q^w(s_t, a_t)$

1.2 Action-Value Gradients

Question 1: Action-Value Gradient Derivation

[1 Point]

Using the chain rule, derive the gradient of the action-value function evaluated at the deterministic policy:

$$\nabla_{\theta} Q^{\mu^k}(s, \mu_{\theta}(s)) = \nabla_{\theta} \mu_{\theta}(s) \nabla_a Q^{\mu^k}(s, a)|_{a=\mu_{\theta}(s)}$$

Answer:

Since θ influences the action-value function exclusively through the chosen action a , the total derivative propagates through the intermediate action variable:

$$\nabla_{\theta} Q^{\mu^k}(s, \mu_{\theta}(s)) = \frac{\partial \mu_{\theta}(s)}{\partial \theta} \cdot \frac{\partial Q^{\mu^k}(s, a)}{\partial a} \Big|_{a=\mu_{\theta}(s)}$$

In standard reinforcement learning notation, $\nabla_{\theta} \mu_{\theta}(s)$ denotes the Jacobian matrix of the policy with respect to θ , and $\nabla_a Q^{\mu^k}(s, a)$ represents the gradient of the Q-function with respect to the action vector. Multiplying these components yields the desired relation:

$$\nabla_{\theta} Q^{\mu^k}(s, \mu_{\theta}(s)) = \nabla_{\theta} \mu_{\theta}(s) \nabla_a Q^{\mu^k}(s, a)|_{a=\mu_{\theta}(s)}$$

1.3 Motivation for Deterministic Policy Gradients

The primary motivation for deterministic policy gradients is that they eliminate the computationally expensive integration over the action space required by stochastic policies. Instead, the gradient can be computed efficiently using only the policy Jacobian from the actor and the action-value gradient from the critic.

1.4 The Deterministic Policy Gradient Theorem

Deterministic Policy Gradient (DPG) Theorem

Consider a deterministic policy $\mu_{\theta} : \mathcal{S} \rightarrow \mathcal{A}$ and the performance objective:

$$J(\mu_{\theta}) = \int_{\mathcal{S}} \rho^{\mu}(s) r(s, \mu_{\theta}(s)) ds = \mathbb{E}_{s \sim \rho^{\mu}} [r(s, \mu_{\theta}(s))]$$

where $\rho^{\mu}(s)$ is the discounted state visitation distribution under policy μ . Then:

$$\nabla_{\theta} J(\mu_{\theta}) = \int_{\mathcal{S}} \rho^{\mu}(s) \nabla_{\theta} \mu_{\theta}(s) \nabla_a Q^{\mu}(s, a)|_{a=\mu_{\theta}(s)} ds$$

or equivalently:

$$\nabla_{\theta} J(\mu_{\theta}) = \mathbb{E}_{s \sim \rho^{\mu}} [\nabla_{\theta} \mu_{\theta}(s) \nabla_a Q^{\mu}(s, a)|_{a=\mu_{\theta}(s)}]$$

Question 2: Proof of the DPG Theorem

[8 Points]

Prove the Deterministic Policy Gradient Theorem by following these steps:

Step 1 Express $J(\mu_\theta)$ in terms of the state-value function $V^{\mu_\theta}(s)$. Show that:

$$J(\mu_\theta) = \int_{\mathcal{S}} \rho^{\mu_\theta}(s_0) V^{\mu_\theta}(s_0) ds_0$$

where $\rho^{\mu_\theta}(s_0)$ is the initial state distribution .

Step 2 Using the Bellman equation for V^{μ_θ} show that:

$$\nabla_\theta V^{\mu_\theta}(s) = \nabla_\theta Q^{\mu_\theta}(s, \mu_\theta(s))$$

Step 3 Expand $\nabla_\theta Q^{\mu_\theta}(s, \mu_\theta(s))$ using the Bellman expectation equation:

$$Q^{\mu_\theta}(s, a) = r(s, a) + \gamma \int_{\mathcal{S}} p(s'|s, a) V^{\mu_\theta}(s') ds'$$

Show that applying the multivariate chain rule yields:

$$\nabla_\theta Q^{\mu_\theta}(s, \mu_\theta(s)) = \nabla_\theta \mu_\theta(s) \nabla_a Q^{\mu_\theta}(s, a) \Big|_{a=\mu_\theta(s)} + \gamma \int_{\mathcal{S}} p(s'|s, \mu_\theta(s)) \nabla_\theta V^{\mu_\theta}(s') ds'$$

Step 4 Apply the integration chain rule recursively across an infinite horizon to show:

$$\nabla_\theta V^{\mu_\theta}(s) = \int_{\mathcal{S}} \sum_{t=0}^{\infty} \gamma^t p(s \rightarrow s', t, \mu_\theta) \nabla_\theta \mu_\theta(s') \nabla_a Q^{\mu_\theta}(s', a) \Big|_{a=\mu_\theta(s')} ds'$$

where $p(s \rightarrow s', t, \mu_\theta)$ is the probability of transitioning from s to s' in t steps under policy μ_θ .

Answer:

To prove the theorem, we construct a continuous analytical narrative by expanding the objective function, taking its total derivative, and unrolling the resulting Bellman recurrence relation:

Step 1: Objective Formulation We begin by expressing the performance objective $J(\mu_\theta)$ in terms of the state-value function $V^{\mu_\theta}(s)$. By substituting the definition of the discounted state visitation distribution $\rho^{\mu_\theta}(s) = \sum_{t=0}^{\infty} \gamma^t \int_{\mathcal{S}} \rho^{\mu_\theta}(s_0) p(s_0 \rightarrow s, t, \mu_\theta) ds_0$ into the primary objective $J(\mu_\theta) = \int_{\mathcal{S}} \rho^{\mu_\theta}(s) r(s, \mu_\theta(s)) ds$, and applying Fubini's theorem to swap the order of integration, we naturally isolate the initial state distribution $\rho^{\mu_\theta}(s_0)$:

$$\begin{aligned} J(\mu_\theta) &= \int_{\mathcal{S}} \left[\sum_{t=0}^{\infty} \gamma^t \int_{\mathcal{S}} \rho^{\mu_\theta}(s_0) p(s_0 \rightarrow s, t, \mu_\theta) ds_0 \right] r(s, \mu_\theta(s)) ds \\ &= \int_{\mathcal{S}} \rho^{\mu_\theta}(s_0) \underbrace{\left[\int_{\mathcal{S}} \sum_{t=0}^{\infty} \gamma^t p(s_0 \rightarrow s, t, \mu_\theta) r(s, \mu_\theta(s)) ds \right]}_{V^{\mu_\theta}(s_0)} ds_0 = \int_{\mathcal{S}} \rho^{\mu_\theta}(s_0) V^{\mu_\theta}(s_0) ds_0 \end{aligned}$$

Step 2 & 3: Policy Gradient Expansion via Chain Rule Since $\mu_\theta(s)$ is strictly deterministic, the state-value matches the action-value evaluated at that single point: $V^{\mu_\theta}(s) = Q^{\mu_\theta}(s, \mu_\theta(s))$. Taking the total gradient with respect to θ requires expanding the Bellman expectation equation $Q^{\mu_\theta}(s, \mu_\theta(s)) = r(s, \mu_\theta(s)) + \gamma \int_{\mathcal{S}} p(s'|s, \mu_\theta(s)) V^{\mu_\theta}(s') ds'$. Applying the multivariate chain rule to account for both explicit action-dependencies and implicit downstream value-dependencies yields:

$$\begin{aligned}\nabla_\theta V^{\mu_\theta}(s) &= \nabla_\theta Q^{\mu_\theta}(s, \mu_\theta(s)) \\ &= \nabla_\theta \mu_\theta(s) \nabla_a Q^{\mu_\theta}(s, a) \Big|_{a=\mu_\theta(s)} + \gamma \int_{\mathcal{S}} p(s'|s, \mu_\theta(s)) \nabla_\theta V^{\mu_\theta}(s') ds'\end{aligned}$$

Step 4: Recursive Unrolling across Infinite Horizon To compact the notation, let $G(s) = \nabla_\theta \mu_\theta(s) \nabla_a Q^{\mu_\theta}(s, a) \Big|_{a=\mu_\theta(s)}$. The expression obtained above forms a linear recurrence relation $\nabla_\theta V^{\mu_\theta}(s) = G(s) + \gamma \int_{\mathcal{S}} p(s \rightarrow s', 1, \mu_\theta) \nabla_\theta V^{\mu_\theta}(s') ds'$. Unrolling this equation repeatedly by substituting the downstream state gradients ($\nabla_\theta V^{\mu_\theta}(s')$) develops a cascading geometric series:

$$\begin{aligned}\nabla_\theta V^{\mu_\theta}(s) &= G(s) + \gamma \int_{\mathcal{S}} p(s \rightarrow s', 1, \mu_\theta) G(s') ds' + \gamma^2 \int_{\mathcal{S}} p(s \rightarrow s'', 2, \mu_\theta) G(s'') ds'' + \dots \\ &= \int_{\mathcal{S}} \sum_{t=0}^{\infty} \gamma^t p(s \rightarrow s', t, \mu_\theta) \nabla_\theta \mu_\theta(s') \nabla_a Q^{\mu_\theta}(s', a) \Big|_{a=\mu_\theta(s')} ds'\end{aligned}$$

Finally, integrating this local state gradient over the initial environment boundaries $\rho^{\mu_\theta}(s_0)$ according to Step 1 transforms the initial distribution into the full discounted state visitation measure $\rho^{\mu_\theta}(s)$, completing the proof:

$$\nabla_\theta J(\mu_\theta) = \int_{\mathcal{S}} \rho^{\mu_\theta}(s) \nabla_\theta \mu_\theta(s) \nabla_a Q^{\mu_\theta}(s, a) \Big|_{a=\mu_\theta(s)} ds = \mathbb{E}_{s \sim \rho^\mu} [\nabla_\theta \mu_\theta(s) \nabla_a Q^\mu(s, a) \Big|_{a=\mu_\theta(s)}]$$

1.5 Benefits of Deterministic Policy Gradients

The Deterministic Policy Gradient (DPG) framework offers two fundamental advantages over the Stochastic Policy Gradient (SPG):

1. **Computational Efficiency:** While SPG requires integrating over the action space and sampling multiple actions per state to approximate the gradient $\nabla_\theta J(\theta) = \mathbb{E}_{s,a} [\nabla_\theta \log \pi_\theta(a|s) Q^\pi(s, a)]$, DPG evaluates only a single deterministic action per state via $\nabla_\theta J(\mu_\theta) = \mathbb{E}_s [\nabla_\theta \mu_\theta(s) \nabla_a Q^\mu(s, a) \Big|_{a=\mu_\theta(s)}]$. This computational saving is especially critical in high-dimensional continuous action spaces.
2. **Limiting Case Relationship:** DPG is mathematically proven to be the limiting case of SPG. For a Gaussian policy $\pi_{\mu_\theta, \sigma}(a|s) = \mathcal{N}(a; \mu_\theta(s), \sigma^2 I)$, as the policy variance approaches zero ($\sigma \rightarrow 0$), the stochastic policy gradient converges exactly to the deterministic policy gradient:

$$\lim_{\sigma \rightarrow 0} \nabla_\theta J(\pi_{\mu_\theta, \sigma}) = \nabla_\theta J(\mu_\theta)$$

1.6 Deterministic Actor-Critic Algorithms

Question 3: Deterministic On-Policy Actor-Critic

[2 Points]

Derive the on-policy deterministic actor-critic algorithm. Assume the actor is a deterministic policy $\mu_\theta(s)$, and the critic is a function approximator $Q^w(s, a) \approx Q^\mu(s, a)$. Transitions are

generated by following μ_θ with exploration noise: $a_t = \mu_\theta(s_t) + \mathcal{N}(0, \sigma^2)$. Provide the update rules for both the critic parameters w and the actor parameters θ .

Answer:

Even though exploratory noise is injected during data collection to ensure state-space coverage ($a_t = \mu_\theta(s_t) + \mathcal{N}(0, \sigma^2)$), the algorithm remains structurally near-on-policy because transitions closely track the target deterministic policy's trajectory. The sample-based update rules are formulated as follows:

- ◇ **Temporal-Difference (TD) Error:** The critic targets the expected return of the clean, unperturbed target policy μ_θ at the next state s_{t+1} :

$$\delta_t = r_t + \gamma Q^w(s_{t+1}, \mu_\theta(s_{t+1})) - Q^w(s_t, a_t)$$

- ◇ **Critic Parameter Update (w):** The weights are updated via semi-gradient descent to minimize the mean squared Bellman error:

$$w_{t+1} = w_t + \alpha_w \delta_t \nabla_w Q^w(s_t, a_t)$$

- ◇ **Actor Parameter Update (θ):** Driven by the Deterministic Policy Gradient (DPG) theorem, the actor updates its parameters along the direction that maximizes the critic's value assessment at the deterministic choice:

$$\theta_{t+1} = \theta_t + \alpha_\theta \nabla_\theta \mu_\theta(s_t) \nabla_a Q^w(s_t, a) \Big|_{a=\mu_\theta(s_t)}$$

Question 4: Deterministic Off-Policy Actor-Critic

[3 Points]

Derive the off-policy deterministic actor-critic algorithm. Assume the target policy is a deterministic policy $\mu_\theta(s)$, the behavior policy is a stochastic policy $\beta(a|s)$, and transitions (s_t, a_t, r_t, s_{t+1}) are sampled from a replay buffer generated by β . Explain analytically why this off-policy deterministic framework does not require action-level importance sampling corrections.

Answer:

In an off-policy deterministic actor-critic framework, environmental interactions are driven by a behavior policy $\beta(a|s)$ to ensure robust exploration, while the objective is to optimize a clean target policy $\mu_\theta(s)$.

1. Algorithmic Update Rules

The sample-based temporal-difference mechanics are formulated as follows:

- ◇ **Off-Policy TD Error:** To ensure the critic evaluates the target policy μ_θ rather than the behavior policy β (akin to Q-learning), the bootstrapping target evaluates the deterministic choice $\mu_\theta(s_{t+1})$ at the next state, discarding the actual logged action a_{t+1} from the buffer:

$$\delta_t = r_t + \gamma Q^w(s_{t+1}, \mu_\theta(s_{t+1})) - Q^w(s_t, a_t)$$

- ◇ **Critic Parameter Update (w):** Minimized via semi-gradient descent using the off-policy TD error:

$$w_{t+1} = w_t + \alpha_w \delta_t \nabla_w Q^w(s_t, a_t)$$

- ◇ **Actor Parameter Update (θ):** Updated via the Deterministic Policy Gradient (DPG) theorem along the state distribution sampled from the replay buffer:

$$\theta_{t+1} = \theta_t + \alpha_\theta \nabla_\theta \mu_\theta(s_t) \nabla_a Q^w(s_t, a) \Big|_{a=\mu_\theta(s_t)}$$

2. Mathematical Proof for Omitting Importance Sampling

In stochastic off-policy policy gradients, the performance objective averages over the action space via an explicit integration:

$$J(\pi_\theta) = \int_{\mathcal{S}} \rho^\beta(s) \int_{\mathcal{A}} \pi_\theta(a|s) Q^{\pi_\theta}(s, a) da ds$$

Differentiating this objective results in $\nabla_\theta \pi_\theta(a|s)$ inside the inner integral. Since the logged actions a_t are drawn from β , converting this inner integral into a computable expectation forces the introduction of the action-level importance sampling ratio $\rho_t = \frac{\pi_\theta(a_t|s_t)}{\beta(a_t|s_t)}$.

Conversely, the deterministic objective function under the behavior state visitation distribution $\rho^\beta(s)$ bypasses action distribution averaging entirely because the policy collapses to a single point-mass mapping $V^{\mu_\theta}(s) = Q^\mu(s, \mu_\theta(s))$:

$$J(\mu_\theta) = \int_{\mathcal{S}} \rho^\beta(s) Q^\mu(s, \mu_\theta(s)) ds$$

Taking the gradient with respect to θ and applying the multivariate chain rule yields:

$$\nabla_\theta J(\mu_\theta) = \int_{\mathcal{S}} \rho^\beta(s) \left(\nabla_\theta \mu_\theta(s) \cdot \nabla_a Q^\mu(s, a) \Big|_{a=\mu_\theta(s)} \right) ds$$

Converting this spatial integral into an expectation gives:

$$\nabla_\theta J(\mu_\theta) = \mathbb{E}_{s \sim \rho^\beta} \left[\nabla_\theta \mu_\theta(s) \cdot \nabla_a Q^\mu(s, a) \Big|_{a=\mu_\theta(s)} \right]$$

Because the expected value is calculated **strictly over the state distribution** $\rho^\beta(s)$, and the action variable evaluated inside the gradient gradient is the actor's own current deterministic proposal ($\mu_\theta(s)$) rather than the historical logged action ($a_t \sim \beta$), **the inner integration over $\mathcal{A}$ completely vanishes**.

Consequently, there is no action distribution mismatch to correct, and the traditional action-level importance sampling ratio is mathematically unnecessary.

Implementation 1: Deterministic Policy Gradient (DPG)

[5 Points]

Implement a near-on-policy deterministic actor-critic baseline for the custom moving-target Reacher environment. The environment code and helper utilities are provided in

`moving_target_reacher_env.py`.

The actor should be updated using the deterministic policy-gradient direction:

$$\nabla_{\theta} J(\mu_{\theta}) \approx \mathbb{E}_s[\nabla_{\theta} \mu_{\theta}(s) \nabla_a Q_{\phi}(s, a)|_{a=\mu_{\theta}(s)}]$$

The critic should be trained with a one-step bootstrapped target:

$$y_t = r_t + \gamma(1 - d_t)Q_{\phi}(s_{t+1}, \mu_{\theta}(s_{t+1})).$$

Since deterministic policies do not explore by themselves, collect data using noisy actions:

$$a_t = \mu_{\theta}(s_t) + \epsilon_t$$

Run the training pipeline, plot the learning curves, and briefly discuss why this baseline is less stable and less sample-efficient than DDPG and TD3.

Answer:

See DPGs.ipynb.

1.7 Compatible Function Approximation

Compatible Function Approximation Theorem for Deterministic Policies

A function approximator $Q^w(s, a)$ is compatible with a deterministic policy $\mu_{\theta}(s)$ if:

1. $\nabla_a Q^w(s, a)|_{a=\mu_{\theta}(s)} = \nabla_{\theta} \mu_{\theta}(s)^{\top} w$
2. The parameters w minimize the mean-squared error of the action-gradients:

$$\epsilon^2 = \mathbb{E}_{s \sim \rho^{\mu}} \left[\left(\nabla_a Q^w(s, a)|_{a=\mu_{\theta}(s)} - \nabla_a Q^{\mu}(s, a)|_{a=\mu_{\theta}(s)} \right)^2 \right]$$

Then, the approximate deterministic policy gradient is unbiased:

$$\mathbb{E}_{s \sim \rho^{\mu}} [\nabla_{\theta} \mu_{\theta}(s) \nabla_a Q^w(s, a)|_{a=\mu_{\theta}(s)}] = \mathbb{E}_{s \sim \rho^{\mu}} [\nabla_{\theta} \mu_{\theta}(s) \nabla_a Q^{\mu}(s, a)|_{a=\mu_{\theta}(s)}]$$

Question 5: Proof of Compatible Function Approximation

[3 Points]

Prove the Compatible Function Approximation theorem by following these steps:

Step 1 State the definition established by Condition 1 for the approximate critic's action gradient.

Step 2 Show that minimizing the action-gradient mean-squared error ϵ^2 with respect to w yields the first-order necessary condition:

$$\mathbb{E}_{s \sim \rho^{\mu}} \left[\nabla_{\theta} \mu_{\theta}(s) \left(\nabla_a Q^w(s, a)|_{a=\mu_{\theta}(s)} - \nabla_a Q^{\mu}(s, a)|_{a=\mu_{\theta}(s)} \right) \right] = 0$$

Step 3 Combine Steps 1 and 2 to prove the unbiased nature of the approximate gradient.

Answer:

The rigorous proof of the Compatible Function Approximation theorem for deterministic policies is derived below:

Step 1

By Condition 1 of the theorem, we directly have the definition that constrains the structural form of the approximate critic's action gradient evaluated at the current policy:

$$\nabla_a Q^w(s, a) \Big|_{a=\mu_\theta(s)} = \nabla_\theta \mu_\theta(s)^\top w$$

Step 2

To minimize the mean-squared error function ϵ^2 with respect to the critic parameters w , we take its partial derivative and set it to zero to establish the first-order necessary condition:

$$\frac{\partial \epsilon^2}{\partial w} = \frac{\partial}{\partial w} \mathbb{E}_{s \sim \rho^\mu} \left[\left(\nabla_a Q^w(s, a) \Big|_{a=\mu_\theta(s)} - \nabla_a Q^\mu(s, a) \Big|_{a=\mu_\theta(s)} \right)^2 \right] = 0$$

Passing the derivative inside the expectation operator and applying the chain rule yields:

$$\mathbb{E}_{s \sim \rho^\mu} \left[2 \cdot \left(\nabla_a Q^w(s, a) \Big|_{a=\mu_\theta(s)} - \nabla_a Q^\mu(s, a) \Big|_{a=\mu_\theta(s)} \right) \cdot \frac{\partial}{\partial w} \left(\nabla_a Q^w(s, a) \Big|_{a=\mu_\theta(s)} \right) \right] = 0$$

From the linear identity verified in Step 1, the derivative with respect to w is simply the policy Jacobian: $\frac{\partial}{\partial w} (\nabla_a Q^w) = \nabla_\theta \mu_\theta(s)$. Substituting this back and eliminating the constant factor gives the orthogonality condition:

$$\mathbb{E}_{s \sim \rho^\mu} \left[\nabla_\theta \mu_\theta(s) \left(\nabla_a Q^w(s, a) \Big|_{a=\mu_\theta(s)} - \nabla_a Q^\mu(s, a) \Big|_{a=\mu_\theta(s)} \right) \right] = 0$$

Step 3

By expanding and distributing the expectation operator over the linear difference relation obtained in Step 2, we get:

$$\mathbb{E}_{s \sim \rho^\mu} \left[\nabla_\theta \mu_\theta(s) \nabla_a Q^w(s, a) \Big|_{a=\mu_\theta(s)} \right] - \mathbb{E}_{s \sim \rho^\mu} \left[\nabla_\theta \mu_\theta(s) \nabla_a Q^\mu(s, a) \Big|_{a=\mu_\theta(s)} \right] = 0$$

Rearranging the terms immediately yields the final objective equality:

$$\mathbb{E}_{s \sim \rho^\mu} [\nabla_\theta \mu_\theta(s) \nabla_a Q^w(s, a) \Big|_{a=\mu_\theta(s)}] = \mathbb{E}_{s \sim \rho^\mu} [\nabla_\theta \mu_\theta(s) \nabla_a Q^\mu(s, a) \Big|_{a=\mu_\theta(s)}]$$

This mathematically proves that substituting the compatible approximate critic Q^w into the policy gradient leaves the resulting actor updates completely unbiased.

1.8 From DPG to DDPG: Practical Challenges

Although the DPG theorem establishes a solid theoretical framework , scaling it to high-dimensional continuous spaces using deep neural networks introduces three critical practical challenges :

- **Sample Inefficiency:** As an on-policy approach, vanilla DPG requires generating entirely new transitions after every single policy update . This continuous data collection process is highly inefficient when environmental interactions are expensive , policy convergence is slow , or state-action spaces are high-dimensional .
- **Deep Network Instability:** Integrating temporal-difference (TD) learning with deep architectures is inherently unstable due to three factors :
 - *Correlated Experience:* Successive environmental transitions are highly correlated, violating the independent and identically distributed (IID) assumption required for stable neural network training .
 - *Moving Targets:* The temporal-difference target $r + \gamma Q^w(s', \mu_\theta(s'))$ shifts constantly because the critic weights w and actor weights θ are updated simultaneously .
 - *Overestimation Bias:* Function approximation errors cause the critic to overestimate action values, which the actor exploits, leading to suboptimal policies .
- **Lack of Exploration:** Because a deterministic policy lacks inherent stochasticity, it constantly selects identical actions for given states . Without an explicit exploration mechanism, the agent easily gets trapped in local optima and overfits to narrow regions of the state space .

1.9 Deep Deterministic Policy Gradient (DDPG)

To scale the Deterministic Policy Gradient framework to complex domains with deep neural networks, the DDPG algorithm incorporates four stabilizing architectural techniques :

- **Experience Replay Buffer:** Transitions (s_t, a_t, r_t, s_{t+1}) are stored in a replay dataset $\mathcal{D}$. Sampling uniform random minibatches breaks the temporal correlation of sequential trajectories, effectively stabilizing value function updates .
- **Target Networks with Soft Updates:** DDPG maintains distinct target networks $Q^{w'}$ and $\mu_{\theta'}$ to construct stable temporal-difference targets: $y_t = r_t + \gamma Q^{w'}(s_{t+1}, \mu_{\theta'}(s_{t+1}))$. The weights are slowly moved toward the online weights via polyak tracking where $\tau \ll 1$:

$$w' \leftarrow \tau w + (1 - \tau)w', \quad \theta' \leftarrow \tau \theta + (1 - \tau)\theta'$$

- **Batch Normalization:** This normalization technique is applied across state dimensions and intermediate layers of both the actor and critic networks to handle varying feature scales gracefully .
- **Exploration Noise:** To enable exploratory behaviors under a deterministic mapping, a temporally correlated Ornstein-Uhlenbeck process noise $\mathcal{N}_t$ is injected directly into the selected action sequence during training: $a_t = \mu_\theta(s_t) + \mathcal{N}_t$.

For each DDPG component, explain how it addresses the practical challenges identified in Section 1.8: (a) Experience Replay Buffer, (b) Target Networks with Soft Updates, (c) Batch Normalization, and (d) Ornstein-Uhlenbeck Exploration Noise. For each component, also mention one limitation or possible failure mode.

Answer:

The analytical evaluation of the four core DDPG architectural components, mapping their theoretical solutions to practical limitations, is detailed below:

(a) Experience Replay Buffer:

- ◇ *Challenge Addressed:* By storing transitions and sampling uniform random minibatches, it breaks the severe temporal correlation of sequential data, shifting the training distribution closer to the IID assumption. It also allows off-policy reuse of historical data, which drastically enhances sample efficiency.
- ◇ *Failure Mode / Limitation:* Uniform random sampling treats all experiences equally. Rare, high-variance, or critically informative transitions (e.g., sparse rewards) can be starved of training updates, requiring extensions like Prioritized Experience Replay (PER).

(b) Target Networks with Soft Updates:

- ◇ *Challenge Addressed:* It directly counteracts the moving target problem in deep TD learning. By updating the target parameters slowly ($\tau \ll 1$), the bootstrapping source y_t remains quasi-static, providing a stable regression objective for the online critic.
- ◇ *Failure Mode / Limitation:* While it stabilizes training, it does **not** mitigate function approximation overestimation bias. The critic still accumulates systematic positive errors that the actor aggressively exploits, which can eventually destabilize the policy landscape without a clipped double Q-learning guardrail.

(c) Batch Normalization:

- ◇ *Challenge Addressed:* It handles varying and highly disparate feature scales across state dimensions (e.g., mixing small radian angles with high rotational motor velocities) by forcing internal layers to maintain stabilized, normalized distributions.
- ◇ *Failure Mode / Limitation:* RL environments are highly non-stationary due to evolving policies. The moving average statistics (μ, σ) tracked during training frequently fail to adapt to abrupt shifts in the state visitation distribution, often inducing sudden, catastrophic policy degradation during deployment when exploration noise is disabled.

(d) Ornstein-Uhlenbeck Exploration Noise:

- ◇ *Challenge Addressed:* It infuses the non-exploratory deterministic policy with continuous, temporally correlated random walk variables. This ensures the generated exploration actions maintain a physical momentum/torque appropriate for continuous robotic controllers, preventing self-canceling high-frequency jitter.

- ◇ *Failure Mode / Limitation:* It introduces severe hyperparameter sensitivity regarding its mean-reversion rates and variance bounds. If annealed/decayed too quickly, the agent terminates exploration prematurely and traps itself in local optima; if decayed too slowly, it destabilizes asymptotic policy convergence during later stages of training.

Implementation 2: Deep Deterministic Policy Gradient (DDPG)

[5 Points]

Extend your DPG implementation to the off-policy DDPG algorithm on the custom moving-target Reacher environment.

The critic should be trained using samples from the replay buffer and the target-network bootstrapped target:

$$y_t = r_t + \gamma(1 - d_t)Q_{\bar{\phi}}(s_{t+1}, \mu_{\bar{\theta}}(s_{t+1}));$$

where $Q_{\bar{\phi}}$ and $\mu_{\bar{\theta}}$ are slowly updated target networks.

The critic loss is

$$L_Q(\phi) = \mathbb{E}_{(s,a,r,s',d) \sim \mathcal{D}}[(Q_{\phi}(s, a) - y)^2].$$

The actor should be updated using the deterministic policy-gradient objective:

$$J(\theta) = \mathbb{E}_{s \sim \mathcal{D}}[Q_{\phi}(s, \mu_{\theta}(s))],$$

or equivalently by minimizing

$$L_{\pi}(\theta) = -\mathbb{E}_{s \sim \mathcal{D}}[Q_{\phi}(s, \mu_{\theta}(s))].$$

During data collection, use noisy actions

$$a_t = \mu_{\theta}(s_t) + \epsilon_t$$

to ensure exploration, but evaluate the final policy without exploration noise.

Run the training pipeline, plot the learning curves, and compare DDPG against your DPG baseline.

In your report, explain how replay buffers and target networks address the sample-efficiency and moving-target instability problems of the near-on-policy DPG baseline.

Answer:

See DPGs.ipynb.

1.10 Function Approximation Error in Deterministic Actor-Critic

Consider a deterministic policy $\mu_{\theta} : \mathcal{S} \rightarrow \mathcal{A}$ and a learned critic $Q_{\phi}(s, a)$. In standard deterministic actor-critic methods, the actor is updated by taking the expected directional derivative of the critic:

$$\nabla_{\theta} J(\theta) \approx \mathbb{E}_{s \sim \rho^{\mu}}[\nabla_{\theta} \mu_{\theta}(s) \nabla_a Q_{\phi}(s, a)|_{a=\mu_{\theta}(s)}]$$

However, when Q_{ϕ} accumulates function approximation errors, the resulting actor gradient becomes systematically biased, severely degrading learning stability.

Question 7: Critic Gradient Error and Actor Bias**[2 Points]**

Assume that for every state $s \in \mathcal{S}$, the critic's action-gradient error is uniformly bounded by $\|\nabla_a Q_\phi(s, a)|_{a=\mu_\theta(s)} - \nabla_a Q^\mu(s, a)|_{a=\mu_\theta(s)}\|_2 \leq \epsilon$, and the operator norm of the policy Jacobian satisfies $\|\nabla_\theta \mu_\theta(s)\|_{op} \leq M$. Prove that the total actor-gradient error satisfies:

$$\left\| \mathbb{E}_{s \sim \rho^\mu} \left[\nabla_\theta \mu_\theta(s) (\nabla_a Q_\phi(s, a) - \nabla_a Q^\mu(s, a)) \Big|_{a=\mu_\theta(s)} \right] \right\|_{Pap_2} \leq M\epsilon$$

Explain analytically why this relationship makes deterministic actor-critic methods especially sensitive to critic approximation errors.

Answer:

Let $\Delta_a(s) = \nabla_a Q_\phi(s, a)|_{a=\mu_\theta(s)} - \nabla_a Q^\mu(s, a)|_{a=\mu_\theta(s)}$ denote the localized action-gradient error vector. By applying Jensen's inequality to move the norm operator inside the expectation, and subsequently invoking the fundamental property of matrix operator norms ($\|Ax\|_2 \leq \|A\|_{op}\|x\|_2$), we bound the error propagation:

$$\begin{aligned} \|\mathbb{E}_{s \sim \rho^\mu} [\nabla_\theta \mu_\theta(s) \Delta_a(s)]\|_2 &\leq \mathbb{E}_{s \sim \rho^\mu} [\|\nabla_\theta \mu_\theta(s) \Delta_a(s)\|_2] \quad (\text{via Jensen's Inequality}) \\ &\leq \mathbb{E}_{s \sim \rho^\mu} [\|\nabla_\theta \mu_\theta(s)\|_{op} \cdot \|\Delta_a(s)\|_2] \quad (\text{via Operator Norm Property}) \end{aligned}$$

Substituting the uniform bounds $\|\nabla_\theta \mu_\theta(s)\|_{op} \leq M$ and $\|\Delta_a(s)\|_2 \leq \epsilon$ into the integrand directly yields the final upper bound:

$$\mathbb{E}_{s \sim \rho^\mu} [M \cdot \epsilon] = M\epsilon$$

Deterministic actor-critic frameworks are exceptionally vulnerable to critic inaccuracies, given that their policy update rule relies exclusively on a pointwise directional gradient computed at a singular action coordinate, $a = \mu_\theta(s)$. While stochastic policy gradients integrate across the entire action manifold—thereby inherently smoothing out localized estimation errors through the expectation of log-probabilities—a deterministic actor directly exploits structural imperfections or local overestimations on the critic's value surface. Consequently, any minor functional error ϵ undergoes linear amplification by the policy's flexibility factor M , aggressively driving the policy parameters toward suboptimal local extrema.

1.11 TD3: Addressing Function Approximation Error

Twin Delayed Deep Deterministic Policy Gradient (TD3) builds on DDPG and introduces three mechanisms to reduce overestimation bias and improve stability: clipped double Q-learning, delayed policy updates, and target policy smoothing. In standard actor-critic methods, the policy is updated using the critic's action-gradient. However, when the critic overestimates the true Q-values, the policy gradient becomes systematically biased, which can lead to suboptimal policies and severe training instability.

Question 8: Overestimation and Policy Performance**[2 Points]**

Explain intuitively why overestimation in the critic leads to poor policy performance. What happens when the actor exploits overestimated Q-values?

Answer:

The actor parameters are optimized via gradient ascent to maximize the values estimated by the critic network. When the critic exhibits function approximation errors that result in the overestimation of certain state-action pairs, these suboptimal actions are incorrectly flagged as highly valuable.

The actor subsequently exploits these spurious peaks on the critic's value surface by shifting its deterministic policy distribution toward these low-reward actions. This creates a destructive positive feedback loop: the actor continuously selects actions based on inflated expectations, and the critic continues to bootstrap from these biased estimates, ultimately degrading asymptotic performance and leading to policy divergence.

Question 9: Sources of Overestimation Bias

[2 Points]

List and briefly explain three sources of overestimation bias in value-based reinforcement learning with function approximation.

Answer:

Three fundamental sources that inherently induce overestimation bias in value-based reinforcement learning with function approximation are detailed below:

1. **The Maximization Operation (Maximization Bias):** Whether using the explicit $\max_a$ operator in discrete Q-learning or the implicit maximization driven by the actor's objective in continuous actor-critic setups, selecting the maximum over noisy estimates introduces a systematic positive bias. If action-value estimates contain zero-mean random noise, the expectation of the maximum value will mathematically overestimate the maximum of the true expectation: $\mathbb{E}[\max_a X_a] \geq \max_a \mathbb{E}[X_a]$.
2. **Function Approximation Inaccuracy:** Deep neural networks have finite capacity and suffer from estimation noise due to non-linear function approximations, limited data samples in unexplored regions, or highly complex reward landscapes. This localized function approximation noise yields structural errors that are systematically selected and accumulated by the policy optimization updates.
3. **Error Propagation via Bootstrapping:** Temporal-difference learning relies on updates that bootstrap from the agent's own downstream value estimates ($r + \gamma \max_{a'} Q(s', a')$). If the critic network overestimates a value at state s' , this positive error is recursively injected into the preceding state value updates during backward passes, accumulating and magnifying the overestimation bias across extended temporal horizons.

1.12 Clipped Double Q-Learning

To address the persistent function approximation error, TD3 utilizes two independent critic networks, Q_{ϕ_1} and Q_{ϕ_2} , and forms a bounded temporal-difference target:

$$y = r + \gamma \min_{i=1,2} Q_{\phi_i}(s', \mu_{\theta'}(s'))$$

Question 10: Minimum of Two Q-Estimates**[3 Points]**

Assume $Q_{\phi_i}(s, a) = Q^*(s, a) + \epsilon_i(s, a)$ for $i = 1, 2$, where ϵ_1 and ϵ_2 are independent, zero-mean, and symmetrically distributed noise variables. Show that:

$$\mathbb{E} \left[\min_{i=1,2} Q_{\phi_i}(s, a) \right] \leq Q^*(s, a)$$

Answer:

We can analytically evaluate the expectation of the minimum operation by invoking the algebraic identity $\min(X, Y) = \frac{X+Y-|X-Y|}{2}$. Let $X = \epsilon_1(s, a)$ and $Y = \epsilon_2(s, a)$ represent the localized estimation noise for each critic. Substituting the critic formulations into the expected minimum equation yields:

$$\begin{aligned} \mathbb{E} \left[\min_{i=1,2} Q_{\phi_i}(s, a) \right] &= \mathbb{E} [\min (Q^*(s, a) + \epsilon_1(s, a), Q^*(s, a) + \epsilon_2(s, a))] \\ &= Q^*(s, a) + \mathbb{E} [\min(\epsilon_1(s, a), \epsilon_2(s, a))] \end{aligned}$$

Applying the algebraic identity directly to the inner noise expectation component expands the relation to:

$$\mathbb{E} [\min(\epsilon_1(s, a), \epsilon_2(s, a))] = \frac{\mathbb{E}[\epsilon_1(s, a)] + \mathbb{E}[\epsilon_2(s, a)] - \mathbb{E}[|\epsilon_1(s, a) - \epsilon_2(s, a)|]}{2}$$

Since both noise processes are strictly zero-mean ($\mathbb{E}[\epsilon_1(s, a)] = \mathbb{E}[\epsilon_2(s, a)] = 0$), the first two terms vanish, isolating the absolute difference:

$$\mathbb{E} \left[\min_{i=1,2} Q_{\phi_i}(s, a) \right] = Q^*(s, a) - \frac{1}{2} \mathbb{E} [|\epsilon_1(s, a) - \epsilon_2(s, a)|]$$

Because the absolute value operator guarantees that $|\epsilon_1(s, a) - \epsilon_2(s, a)| \geq 0$ across all samples, its expected value must also be non-negative ($\mathbb{E}[\cdot] \geq 0$). Thus, subtracting this non-negative quantity guarantees the upper-bound condition:

$$\mathbb{E} \left[\min_{i=1,2} Q_{\phi_i}(s, a) \right] \leq Q^*(s, a)$$

This confirms that the clipped double Q-learning mechanism mathematically suppresses overestimation bias, converting it into a manageable underestimation framework.

Question 11: Underestimation vs. Overestimation**[2 Points]**

Explain the trade-off: why might a slight underestimation bias be less harmful than overestimation bias in actor-critic methods?

Answer:

The fundamental asymmetry in harm between these two biases lies in how the actor's optimization gradient reacts to the critic's function surface:

- ◇ **The Danger of Overestimation Bias:** When a critic overestimates the value of an action, the actor's gradient updates aggressively drive the deterministic policy toward that specific action. This sets off a destructive positive feedback loop: the actor blindly commits to a suboptimal action due to inflated expectations, and the critic continuously propagates this positive error backward via temporal-difference bootstrapping, frequently leading to catastrophic policy divergence.
- ◇ **The Safety of Underestimation Bias:** Conversely, a slight underestimation bias induces a form of systematic "pessimism" or caution. If the critic underestimates certain action values, the actor simply avoids prioritizing those choices. Crucially, this does **not** trigger a runaway feedback loop; an underestimated action will not be repeatedly selected by the actor, meaning its negative error cannot easily propagate or compound through the bootstrapping chain.

While severe underestimation may marginally slow down initial exploration, it naturally prevents structural exploitation loops, maintaining stable asymptotic policy convergence.

1.13 Delayed Policy Updates

Question 12: Delayed Policy Updates

[2 Points]

Explain why updating the policy too frequently relative to the critic can be harmful. Interpret delayed policy updates as an approximate two-time-scale method.

Answer:

The analytical breakdown of this temporal asymmetry highlights two key systemic dynamics:

1.Harm of Premature Updates Updating the actor and critic concurrently ($d = 1$) forces the policy to optimize against an un-converged, volatile value surface plagued by localized estimation noise and micro-variances. The actor directly absorbs these transient critic errors into its parameters, inducing severe policy oscillations and cognitive "confusion" that often leads to training divergence.

2.Two-Time-Scale Interpretation This framework perfectly mirrors a two-time-scale stochastic approximation scheme:

- ◇ **Fast-Scale (Critic):** Updates at every step to rapidly minimize the Bellman residual, tracking values under a quasi-static policy.
- ◇ **Slow-Scale (Actor):** Updates under a delay $\mathcal{O}(1/d)$, ensuring parameter changes occur only after directional action-gradients have securely converged to trustworthy orientations.

1.14 Target Policy Smoothing

Question 13: Target Policy Smoothing Intuition

[2 Points]

Explain the intuition behind target policy smoothing. Why does adding clipped noise to the target action help?

Answer:

Deterministic neural network critics are highly susceptible to overfitting along narrow, high-frequency value flares on their function surfaces. Target policy smoothing forces the critic to learn a robust and smooth expectation over a localized continuous space around the target action, rather than evaluating a single deterministic point.

Adding noise serves as a structural regularizer that smooths the value landscape, ensuring that similar actions yield similar value expectations. Clipping the noise is a crucial practical constraint: it keeps the perturbed action within a realistic neighborhood of the true target policy distribution, preventing the bootstrapping target from incorporating completely anomalous or highly out-of-bounds action values.

Question 14: Target Smoothing and Regularization

[3 Points]

Ignoring clipping, suppose $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$ and $Q(s, a)$ is twice differentiable in a . Derive a second-order approximation for

$$\mathbb{E}_\epsilon[Q(s', \mu(s') + \epsilon)].$$

Explain how this relates to local regularization of the critic target.

Answer:

Let $a' = \mu(s')$ denote the deterministic target action. To derive the approximation, we apply a second-order multivariate Taylor series expansion to the action-value function $Q(s', a' + \epsilon)$ around a' , and then take the expectation with respect to the noise distribution $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$:

$$\begin{aligned} \mathbb{E}_\epsilon[Q(s', a' + \epsilon)] &\approx \mathbb{E}_\epsilon \left[Q(s', a') + \nabla_a Q(s', a')^\top \epsilon + \frac{1}{2} \epsilon^\top \nabla_a^2 Q(s', a') \epsilon \right] \\ &= Q(s', a') + \nabla_a Q(s', a')^\top \underbrace{\mathbb{E}_\epsilon[\epsilon]}_0 + \frac{1}{2} \mathbb{E}_\epsilon \left[\text{Tr} \left(\epsilon^\top \nabla_a^2 Q(s', a') \epsilon \right) \right] \\ &= Q(s', a') + 0 + \frac{1}{2} \text{Tr} \left(\nabla_a^2 Q(s', a') \cdot \underbrace{\mathbb{E}_\epsilon[\epsilon \epsilon^\top]}_{\sigma^2 I} \right) \\ &= Q(s', \mu(s')) + \frac{\sigma^2}{2} \text{Tr} (\nabla_a^2 Q(s', \mu(s'))) \end{aligned}$$

Connection to Local Regularization: The derived second-order modification term $\text{Tr} (\nabla_a^2 Q(s', \mu(s')))$ represents the Laplacian operator ($\Delta_a Q$), which sums the unmixed second partial derivatives and directly gauges the local curvature (sharpness) of the critic's function surface. Substituting this expected value into the temporal-difference objective implicitly adds a curvature penalty to the optimization landscape. This acts as a **local Laplacian regularizer** that prevents the critic from fitting sharp, over-optimistic value flares in the immediate neighborhood of the target action, effectively forcing the neural network to learn a smooth and generalized value topology.

Extend your DDPG implementation to TD3 on the custom moving-target Reacher environment. TD3 uses two critics Q_{ϕ_1} and Q_{ϕ_2} to reduce overestimation bias. The critic target should be

$$y_t = r_t + \gamma(1 - d_t) \min_{i=1,2} Q_{\phi_i}^-(s_{t+1}, \mu_{\bar{\theta}}(s_{t+1}) + \epsilon),$$

where

$$\epsilon \sim \text{clip}(\mathcal{N}(0, \sigma^2), -c, c).$$

The perturbed target action should also be clipped to the valid action range of the environment. The critic loss is

$$L_Q(\phi_1, \phi_2) = \mathbb{E}_{(s,a,r,s',d) \sim \mathcal{D}} [(Q_{\phi_1}(s, a) - y)^2 + (Q_{\phi_2}(s, a) - y)^2].$$

The actor should be updated less frequently than the critics. When an actor update is performed, use

$$L_{\pi}(\theta) = -\mathbb{E}_{s \sim \mathcal{D}} [Q_{\phi_1}(s, \mu_{\theta}(s))].$$

Implement the three main TD3 modifications:

1. **Clipped double Q-learning:** use $\min(Q_{\phi_1}^-, Q_{\phi_2}^-)$ in the target.
2. **Delayed policy updates:** update the actor and target networks once every d critic updates.
3. **Target policy smoothing:** add clipped Gaussian noise to the target action.

Run the training pipeline and compare TD3 against DDPG and the near-on-policy DPG baseline. Include learning curves and at least one ablation, such as removing clipped double Q-learning, removing delayed actor updates, or removing target policy smoothing. In your report, explain which TD3 component had the largest effect and relate your observations to critic overestimation and actor exploitation of critic errors.

Answer:

See DPGs.ipynb.

2 Soft Actor-Critic

2.1 Maximum Entropy Framework

The maximum entropy reinforcement learning framework augments the standard reward maximization objective with an entropy regularization term. This formulation structurally incentivizes the policy to maintain stochasticity, thereby enhancing exploration and preventing premature convergence. Consequently, the resulting policy is significantly more resilient to environmental variations and function approximation errors.

Question 1: Maximum Entropy Framework

[3 Points]

Explain intuitively why adding entropy improves exploration and robustness. Your answer should discuss:

1. why entropy prevents premature collapse to deterministic policies;
2. why stochasticity is useful in environments with multiple near-optimal actions;
3. why the maximum-entropy objective can be more robust to approximation error than pure reward maximization.

Answer:

The integration of an entropy bonus into the reinforcement learning objective enhances both exploration and robustness through the following distinct conceptual mechanisms:

- ◇ **Prevention of Premature Policy Collapse:** Intuitively, if an agent becomes overly confident in a specific action, the probability distribution peaks sharply, causing its local entropy to diminish toward zero. By introducing the entropy penalty, the objective discourages such narrow choices. Maintaining higher entropy forces the policy to assign non-zero probabilities to alternative actions, effectively preventing the policy from prematurely collapsing into a sub-optimal deterministic state and ensuring sustained exploration.
- ◇ **Exploitation of Multiple Near-Optimal Actions:** Furthermore, in environments characterized by multiple distinct action paths yielding comparable rewards, a standard reward-maximization policy often converges arbitrarily to a single mode. By maximizing entropy alongside rewards, the policy is regularized to actively distribute its probability mass across all near-optimal actions. Consequently, the agent successfully learns the full spectrum of effective strategies while simultaneously maximizing the overall return. [
- ◇ **Robustness to Function Approximation Errors:** Moreover, function approximation errors within the learned critic typically manifest as sudden, erroneously high spikes in estimated action-values. In a conventional framework, the actor would immediately and aggressively exploit these artifacts, culminating in abrupt, unstable policy updates. Intuitively, the maximum entropy objective acts as a structural smoother; because shifting to a highly localized, deterministic policy incurs a severe entropy penalty, it mitigates sharp adjustments and prevents highly sensitive or erratic changes in response to critic noise.

Question 2: Soft Value Function and the Variational Formula

[5 Points]

Show that the entropy-optimized soft value function can be written as

$$\mathcal{V}_Q(s) = \alpha \log \int_{\mathcal{A}} \exp \left(\frac{1}{\alpha} Q(s, a) \right) da.$$

Also show that the maximizing policy is the Boltzmann policy

$$\pi_Q^*(a|s) = \frac{\exp \left(\frac{1}{\alpha} Q(s, a) \right)}{\int_{\mathcal{A}} \exp \left(\frac{1}{\alpha} Q(s, \bar{a}) \right) d\bar{a}}.$$

Finally, explain why Equation (3) is not generally equal to $V^\pi(s)$ for an arbitrary fixed policy π .

Answer:

To determine the optimal action distribution that maximizes the entropy-regularized value function

given a fixed action-value function $Q(s, a)$, we define a constrained functional optimization problem for a given state s . The performance objective is expressed as:

$$\max_{\pi} \int_{\mathcal{A}} \pi(a|s) Q(s, a) da - \alpha \int_{\mathcal{A}} \pi(a|s) \log \pi(a|s) da$$

subject to the fundamental probability normalization constraint:

$$\int_{\mathcal{A}} \pi(a|s) da = 1.$$

Consequently, we construct the functional Lagrangian expression by incorporating a Lagrange multiplier $\lambda \in \mathbb{R}$:

$$\mathcal{L}(\pi, \lambda) = \int_{\mathcal{A}} \pi(a|s) Q(s, a) da - \alpha \int_{\mathcal{A}} \pi(a|s) \log \pi(a|s) da - \lambda \left(\int_{\mathcal{A}} \pi(a|s) da - 1 \right).$$

To locate the optimal policy $\pi_Q^*(a|s)$, we evaluate the functional derivative of $\mathcal{L}(\pi, \lambda)$ with respect to $\pi(a|s)$ and equate it to zero:

$$\frac{\delta \mathcal{L}}{\delta \pi(a|s)} = Q(s, a) - \alpha (\log \pi(a|s) + 1) - \lambda = 0.$$

Isolating the logarithmic term yields the following intermediate relation:

$$\log \pi(a|s) = \frac{Q(s, a) - \lambda}{\alpha} - 1.$$

By applying the exponential function to both sides of the equation, the structural form of the optimal policy is recovered as:

$$\pi_Q^*(a|s) = \exp \left(\frac{Q(s, a)}{\alpha} \right) \exp \left(-\frac{\lambda}{\alpha} - 1 \right).$$

Furthermore, substituting this expression back into the axiomatic normalization constraint enables the determination of the constant scaling factor:

$$\int_{\mathcal{A}} \exp \left(\frac{Q(s, a)}{\alpha} \right) \exp \left(-\frac{\lambda}{\alpha} - 1 \right) da = 1 \implies \exp \left(-\frac{\lambda}{\alpha} - 1 \right) = \frac{1}{\int_{\mathcal{A}} \exp \left(\frac{Q(s, \bar{a})}{\alpha} \right) d\bar{a}}.$$

Defining the denominator as the partition function $Z(s) = \int_{\mathcal{A}} \exp \left(\frac{1}{\alpha} Q(s, \bar{a}) \right) d\bar{a}$, the maximizing policy structurally reduces to the standard Boltzmann distribution:

$$\pi_Q^*(a|s) = \frac{\exp \left(\frac{1}{\alpha} Q(s, a) \right)}{\int_{\mathcal{A}} \exp \left(\frac{1}{\alpha} Q(s, \bar{a}) \right) d\bar{a}}.$$

To derive the closed-form expression for the soft value function $\mathcal{V}_Q(s)$, we evaluate the original objective under the maximizing policy $\pi_Q^*(a|s)$. Intuitively, taking the natural logarithm of the Boltzmann policy yields:

$$\log \pi_Q^*(a|s) = \frac{1}{\alpha} Q(s, a) - \log Z(s).$$

Thus, the regularized value integrand simplifies line-by-line via the following algebraic expansion:

$$\begin{aligned}
\mathcal{V}_Q(s) &= \int_{\mathcal{A}} \pi_Q^*(a|s) [Q(s, a) - \alpha \log \pi_Q^*(a|s)] da \\
&= \int_{\mathcal{A}} \pi_Q^*(a|s) \left[Q(s, a) - \alpha \left(\frac{1}{\alpha} Q(s, a) - \log Z(s) \right) \right] da \\
&= \int_{\mathcal{A}} \pi_Q^*(a|s) [\alpha \log Z(s)] da \\
&= \alpha \log Z(s) \int_{\mathcal{A}} \pi_Q^*(a|s) da.
\end{aligned}$$

Since the optimal policy integrates to unity, we successfully obtain the log-sum-exp variational formula:

$$\mathcal{V}_Q(s) = \alpha \log Z(s) = \alpha \log \int_{\mathcal{A}} \exp \left(\frac{1}{\alpha} Q(s, a) \right) da.$$

Moreover, concerning the final component of the problem, $\mathcal{V}_Q(s)$ is not generally equivalent to $V^\pi(s)$ for an arbitrary fixed policy π . Mathematically, $\mathcal{V}_Q(s)$ acts as a greedy supremum operator over a decoupled, static $Q(s, a)$ function. Conversely, $V^\pi(s)$ denotes the true soft state-value function evaluated under a specific, potentially sub-optimal policy π , inherently satisfying the soft Bellman expectation equation where actions are sampled strictly according to π rather than the optimal Boltzmann distribution π_Q^* . Consequently, equality is preserved if and only if the arbitrary policy identically matches the maximizing distribution, such that $\pi = \pi_Q^*$.

Question 3: Analysis on Limiting Behavior

[2 Points]

Analyze the limiting behavior as $\alpha \rightarrow 0$ and $\alpha \rightarrow \infty$. Your answer should discuss both the soft value $\alpha \log \int_{\mathcal{A}} \exp \left(\frac{1}{\alpha} Q(s, a) \right) da$ and the Boltzmann policy $\pi_Q^*(a|s)$.

Answer:

The temperature parameter α regulates the trade-off between reward optimization and entropy maximization, yielding distinct asymptotic boundaries.

As $\alpha \rightarrow 0^+$, entropy regularization vanishes, reclaiming classical greedy control. Intuitively, the scaling factor $\frac{1}{\alpha} \rightarrow \infty$ causes the exponential term of the unique global maximum $a^* = \arg \max_{a \in \mathcal{A}} Q(s, a)$ to infinitely dominate the partition function. Consequently, the Boltzmann policy collapses to a Dirac delta distribution, $\lim_{\alpha \rightarrow 0^+} \pi_Q^*(a|s) = \delta(a - a^*)$. Moreover, invoking Laplace's principle demonstrates that the soft value function is entirely dominated by the supremum of the exponent, precisely recovering the standard hard-max operation:

$$\lim_{\alpha \rightarrow 0^+} \mathcal{V}_Q(s) = \max_{a \in \mathcal{A}} Q(s, a).$$

Conversely, as $\alpha \rightarrow \infty$, the objective prioritizes entropy maximization exclusively, rendering environmental rewards negligible. Since $\frac{1}{\alpha} \rightarrow 0$, every exponential term approaches unity, forcing the policy to uniformize completely over a bounded action space, $\lim_{\alpha \rightarrow \infty} \pi_Q^*(a|s) = \frac{1}{|\mathcal{A}|}$. Furthermore, a first-order Taylor expansion of the integrand yields $\int_{\mathcal{A}} \exp \left(\frac{1}{\alpha} Q(s, a) \right) da \approx |\mathcal{A}| + \frac{1}{\alpha} \int_{\mathcal{A}} Q(s, a) da$. Consequently, applying the linear approximation $\log(1 + x) \approx x$ simplifies the variational formula

line-by-line:

$$\begin{aligned} \mathcal{V}_Q(s) &\approx \alpha \log \left[|\mathcal{A}| \left(1 + \frac{1}{\alpha |\mathcal{A}|} \int_{\mathcal{A}} Q(s, a) da \right) \right] \\ &= \alpha \log |\mathcal{A}| + \alpha \log \left(1 + \frac{1}{\alpha |\mathcal{A}|} \int_{\mathcal{A}} Q(s, a) da \right) \\ &\approx \alpha \log |\mathcal{A}| + \mathbb{E}_{a \sim \text{Unif}(\mathcal{A})} [Q(s, a)]. \end{aligned}$$

Thus, as $\alpha \rightarrow \infty$, the linear term $\alpha \log |\mathcal{A}|$ dictates the asymptotic divergence, while the secondary reward-dependent expectation collapses to a simple uniform average.

2.2 Soft Policy Evaluation

Soft Policy Evaluation Lemma

Consider the soft Bellman backup operator $\mathcal{T}^\pi$ in Equation (4). Let $Q^0 : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ be arbitrary. Define $Q^{k+1} = \mathcal{T}^\pi Q^k$. Then Q^k converges to the soft Q -value Q^π as $k \rightarrow \infty$.

Question 4: Soft Policy Evaluation Lemma

[3 Points]

Prove the Soft Policy Evaluation Lemma.

Answer:

To establish that the iterative sequence $Q^{k+1} = \mathcal{T}^\pi Q^k$ uniquely converges to the true soft action-value function Q^π as $k \rightarrow \infty$, it is sufficient to demonstrate that the soft Bellman backup operator $\mathcal{T}^\pi$ is a strict γ -contraction mapping under the supremum norm $\|\cdot\|_\infty$. Let Q_1 and Q_2 denote two arbitrary continuous action-value functions mapping $\mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$. Evaluating the absolute pointwise difference after a single application of the operator yields:

$$|\mathcal{T}^\pi Q_1(s, a) - \mathcal{T}^\pi Q_2(s, a)| = \gamma |\mathbb{E}_{s' \sim p(\cdot|s, a)} [V_{Q_1}^\pi(s') - V_{Q_2}^\pi(s')]|.$$

Furthermore, expanding this expression by substituting the definition of the soft state-value function reveals that the policy-dependent entropy regularization terms cancel out identically:

$$\begin{aligned} V_{Q_1}^\pi(s') - V_{Q_2}^\pi(s') &= \mathbb{E}_{a' \sim \pi} [Q_1(s', a') - \alpha \log \pi(a'|s')] - \mathbb{E}_{a' \sim \pi} [Q_2(s', a') - \alpha \log \pi(a'|s')] \\ &= \mathbb{E}_{a' \sim \pi} [Q_1(s', a') - Q_2(s', a')]. \end{aligned}$$

Applying Jensen's inequality and bounding the absolute difference by the supremum norm across the entire state-action space yields:

$$|V_{Q_1}^\pi(s') - V_{Q_2}^\pi(s')| \leq \mathbb{E}_{a' \sim \pi} [|Q_1(s', a') - Q_2(s', a')|] \leq \mathbb{E}_{a' \sim \pi} [\|Q_1 - Q_2\|_\infty] = \|Q_1 - Q_2\|_\infty.$$

Consequently, substituting this uniform bound back into the primary operator difference equation implies:

$$|\mathcal{T}^\pi Q_1(s, a) - \mathcal{T}^\pi Q_2(s, a)| \leq \gamma \mathbb{E}_{s' \sim p(\cdot|s, a)} [\|Q_1 - Q_2\|_\infty] = \gamma \|Q_1 - Q_2\|_\infty.$$

Taking the supremum of both sides over all possible state-action pairs $(s, a) \in \mathcal{S} \times \mathcal{A}$ formalizes the contraction condition:

$$\|\mathcal{T}^\pi Q_1 - \mathcal{T}^\pi Q_2\|_\infty \leq \gamma \|Q_1 - Q_2\|_\infty.$$

Moreover, since the discount factor satisfies $\gamma \in [0, 1)$, the operator $\mathcal{T}^\pi$ constitutes a strict γ -contraction mapping. Intuitively, by invoking the Banach Fixed-Point Theorem, any such contraction mapping operating on a complete metric space is guaranteed to possess a unique fixed point, which satisfies $Q^\pi = \mathcal{T}^\pi Q^\pi$. Consequently, repeating the soft Bellman backup operation from an arbitrary initial function Q^0 guarantees asymptotic convergence to the unique soft action-value function Q^π , establishing that $\lim_{k \rightarrow \infty} Q^k = Q^\pi$.

Question 5: Soft Bellman Residual Bound

[4 Points]

Let Q^π be the fixed point of $\mathcal{T}^\pi$. Suppose an approximate critic Q satisfies

$$\|\mathcal{T}^\pi Q - Q\|_\infty \leq \epsilon.$$

Prove that

$$\|Q - Q^\pi\|_\infty \leq \frac{\epsilon}{1 - \gamma}.$$

Answer:

By the Soft Policy Evaluation Lemma, the iterative sequence of functions generated by the soft Bellman backup operator converges asymptotically to the unique fixed point, establishing that $\lim_{k \rightarrow \infty} \mathcal{T}^k Q = Q^\pi$. Consequently, the uniform distance between the approximate critic Q and the true soft action-value function Q^π can be rigorously expressed as an infinite telescoping series. Applying the generalized triangle inequality to this formulation yields:

$$\|Q - Q^\pi\|_\infty = \left\| \sum_{t=0}^{\infty} (\mathcal{T}^t Q - \mathcal{T}^{t+1} Q) \right\|_\infty \leq \sum_{t=0}^{\infty} \|\mathcal{T}^t Q - \mathcal{T}^{t+1} Q\|_\infty.$$

Furthermore, a uniform bound on each consecutive difference term within the summation can be derived via mathematical induction. By invoking the strict γ -contraction mapping property of the soft Bellman operator under the supremum norm, it follows that for any baseline index $t \geq 0$:

$$\|\mathcal{T}^{t+1} Q - \mathcal{T}^t Q\|_\infty \leq \gamma^t \|\mathcal{T} Q - Q\|_\infty.$$

Substituting the primary assumption that the immediate one-step soft Bellman residual is uniformly bounded by ϵ simplifies this inductive relation line-by-line:

$$\|\mathcal{T}^{t+1} Q - \mathcal{T}^t Q\|_\infty \leq \gamma^t \epsilon.$$

Consequently, substituting this bound back into the primary infinite summation expression directly factors out the scalar error constant:

$$\|Q - Q^\pi\|_\infty \leq \sum_{t=0}^{\infty} \gamma^t \epsilon = \epsilon \sum_{t=0}^{\infty} \gamma^t.$$

Intuitively, because the standard reinforcement learning discount factor satisfies $\gamma \in [0, 1)$, the resulting infinite geometric series is guaranteed to converge analytically. Evaluating this infinite geometric sum completes the proof:

$$\|Q - Q^\pi\|_\infty \leq \frac{\epsilon}{1 - \gamma}.$$

2.3 Soft Policy Improvement

Soft Policy Improvement Lemma

Let $\pi_{old} \in \Pi$ and define:

$$\pi_{new} = \arg \min_{\pi' \in \Pi} D_{KL} \left(\pi'(\cdot|s_t) \left\| \frac{\exp(\frac{1}{\alpha} Q^{\pi_{old}}(s_t, \cdot))}{Z^{\pi_{old}}(s_t)} \right\| \right)$$

Then $Q^{\pi_{new}}(s_t, a_t) \geq Q^{\pi_{old}}(s_t, a_t)$ for all (s_t, a_t) with $|\mathcal{A}| < \infty$.

Question 6: Soft Policy Improvement Lemma

[3 Points]

Prove the Soft Policy Improvement Lemma.

Answer:

By definition, the updated parametric policy π_{new} is formulated as the minimizer of the Kullback-Leibler (KL) divergence against the energy-based Boltzmann distribution induced by the current soft action-value function $Q^{\pi_{old}}$. Consequently, by the property of information-theoretic optimality, evaluating this objective under π_{new} must yield a value less than or equal to that obtained under π_{old} or any other arbitrary distribution in the policy space:

$$D_{KL} \left(\pi_{new}(\cdot|s_t) \left\| \frac{\exp(\frac{1}{\alpha} Q^{\pi_{old}}(s_t, \cdot))}{Z^{\pi_{old}}(s_t)} \right\| \right) \leq D_{KL} \left(\pi_{old}(\cdot|s_t) \left\| \frac{\exp(\frac{1}{\alpha} Q^{\pi_{old}}(s_t, \cdot))}{Z^{\pi_{old}}(s_t)} \right\| \right).$$

Expanding the KL divergence definition explicitly into its integral expectation components and canceling the shared, action-independent log-partition function $\log Z^{\pi_{old}}(s_t)$ from both sides of the inequality yields:

$$\mathbb{E}_{a \sim \pi_{new}} \left[\log \pi_{new}(a|s_t) - \frac{1}{\alpha} Q^{\pi_{old}}(s_t, a) \right] \leq \mathbb{E}_{a \sim \pi_{old}} \left[\log \pi_{old}(a|s_t) - \frac{1}{\alpha} Q^{\pi_{old}}(s_t, a) \right].$$

Furthermore, multiplying the entire inequality by the negative temperature coefficient $-\alpha$ reverses the directional inequality sign. Intuitively, substituting the axiomatic definition of the original soft state-value function $V^{\pi_{old}}(s_t) = \mathbb{E}_{a \sim \pi_{old}} [Q^{\pi_{old}}(s_t, a) - \alpha \log \pi_{old}(a|s_t)]$ transforms the expression into the fundamental single-step policy improvement bound:

$$\mathbb{E}_{a \sim \pi_{new}} [Q^{\pi_{old}}(s_t, a) - \alpha \log \pi_{new}(a|s_t)] \geq V^{\pi_{old}}(s_t).$$

Next, we map this localized single-step inequality across an extended temporal horizon by invoking the standard soft Bellman expectation equation for the current policy:

$$Q^{\pi_{old}}(s_t, a_t) = r(s_t, a_t) + \gamma \mathbb{E}_{s_{t+1} \sim p} [V^{\pi_{old}}(s_{t+1})].$$

Substituting the single-step improvement bound evaluated at the downstream state s_{t+1} directly into the future state-value expectation term yields the following recursive inequality:

$$Q^{\pi_{old}}(s_t, a_t) \leq r(s_t, a_t) + \gamma \mathbb{E}_{\substack{s_{t+1} \sim p \\ a_{t+1} \sim \pi_{new}}} [Q^{\pi_{old}}(s_{t+1}, a_{t+1}) - \alpha \log \pi_{new}(a_{t+1}|s_{t+1})].$$

Moreover, because the right-hand side of this relation matches the exact application of the soft Bellman backup operator $\mathcal{T}^{\pi_{\text{new}}}$ to the function $Q^{\pi_{\text{old}}}$, the expression simplifies structurally to $Q^{\pi_{\text{old}}} \leq \mathcal{T}^{\pi_{\text{new}}} Q^{\pi_{\text{old}}}$. Since the soft Bellman operator is strictly monotone, we can repeatedly apply $\mathcal{T}^{\pi_{\text{new}}}$ to both sides to construct a monotonically non-decreasing sequence of action-value estimates:

$$Q^{\pi_{\text{old}}} \leq \mathcal{T}^{\pi_{\text{new}}} Q^{\pi_{\text{old}}} \leq (\mathcal{T}^{\pi_{\text{new}}})^2 Q^{\pi_{\text{old}}} \leq \dots \leq \lim_{k \rightarrow \infty} (\mathcal{T}^{\pi_{\text{new}}})^k Q^{\pi_{\text{old}}}.$$

Consequently, invoking the Soft Policy Evaluation Lemma guarantees that this infinite iterative sequence converges uniquely to the fixed point of the target policy, establishing that $\lim_{k \rightarrow \infty} (\mathcal{T}^{\pi_{\text{new}}})^k Q^{\pi_{\text{old}}} = Q^{\pi_{\text{new}}}$. This mathematical convergence completes the proof, confirming a monotonic increase in the soft action-value function across all valid configurations:

$$Q^{\pi_{\text{new}}}(s_t, a_t) \geq Q^{\pi_{\text{old}}}(s_t, a_t), \quad \forall (s_t, a_t) \in \mathcal{S} \times \mathcal{A}.$$

2.4 Soft Policy Iteration

Soft Policy Iteration Theorem

Repeated application of soft policy evaluation and soft policy improvement from any $\pi \in \Pi$ converges to a policy π^ such that $Q^{\pi^*}(s_t, a_t) \geq Q^{\pi}(s_t, a_t)$ for all $\pi \in \Pi$ and $(s_t, a_t) \in \mathcal{S} \times \mathcal{A}$, assuming $|\mathcal{A}| < \infty$.*

Question 7: Soft Policy Iteration Theorem

[3 Points]

Prove the Soft Policy Iteration Theorem.

Answer:

Let π_k denote the policy at iteration k , and let Q^{π_k} be its corresponding soft action-value function computed via soft policy evaluation. By defining the subsequent policy π_{k+1} through the soft policy improvement operator, the Soft Policy Improvement Lemma guarantees a monotonic increase in the value function across all state-action pairs:

$$Q^{\pi_{k+1}}(s_t, a_t) \geq Q^{\pi_k}(s_t, a_t) \quad \forall (s_t, a_t) \in \mathcal{S} \times \mathcal{A}.$$

Consequently, the sequence of functions $\{Q^{\pi_k}\}_{k=0}^{\infty}$ is monotonically non-decreasing pairs. Furthermore, assuming that the environmental reward function $r(s, a)$ is bounded from above and the action space $\mathcal{A}$ is finite, the maximum possible entropy $\mathcal{H}(\pi) \leq \log |\mathcal{A}|$ is likewise bounded. Intuitively, these conditions ensure that the infinite discounted return-plus-entropy objective is strictly bounded from above. By invoking the Monotone Convergence Theorem, the sequence $\{Q^{\pi_k}\}_{k=0}^{\infty}$ must converge to a bounded unique fixed point, denoted as Q^{π^*} . At this fixed-point boundary, the policy satisfies the invariance relation $\pi_k = \pi_{k+1} = \pi^*$, meaning no further information-theoretic improvement can be executed. Thus, the resulting value function matches the optimal soft value function, confirming that $Q^{\pi^*}(s_t, a_t) \geq Q^{\pi}(s_t, a_t)$ for all alternative strategies $\pi \in \Pi$.

2.5 Soft Actor-Critic Losses

Question 8: SAC Actor Objective as KL Minimization

[3 Points]

For a fixed critic Q_ϕ , show that minimizing

$$J_\pi(\theta) = \mathbb{E}_{s \sim \mathcal{D}, a \sim \pi_\theta(\cdot|s)} [\alpha \log \pi_\theta(a|s) - Q_\phi(s, a)]$$

is equivalent, up to a state-dependent constant, to minimizing

$$\mathbb{E}_{s \sim \mathcal{D}} [D_{KL} \left(\pi_\theta(\cdot|s) \parallel \frac{\exp(\frac{1}{\alpha} Q_\phi(s, \cdot))}{Z_\phi(s)} \right)].$$

Explain the connection between this actor update and soft policy improvement.

Answer:

To establish the equivalence of the objectives, we expand the Kullback-Leibler (KL) divergence between the parametric policy $\pi_\theta(\cdot|s)$ and the target energy-based Boltzmann distribution inside the expectation over the replay buffer states $s \sim \mathcal{D}$:

$$D_{KL} \left(\pi_\theta(\cdot|s) \parallel \frac{\exp(\frac{1}{\alpha} Q_\phi(s, \cdot))}{Z_\phi(s)} \right) = \mathbb{E}_{a \sim \pi_\theta(\cdot|s)} \left[\log \pi_\theta(a|s) - \log \left(\frac{\exp(\frac{1}{\alpha} Q_\phi(s, a))}{Z_\phi(s)} \right) \right].$$

Applying logarithmic identities to separate the target distribution fraction yields:

$$= \mathbb{E}_{a \sim \pi_\theta(\cdot|s)} \left[\log \pi_\theta(a|s) - \frac{1}{\alpha} Q_\phi(s, a) + \log Z_\phi(s) \right].$$

Since the partition function $Z_\phi(s) = \int_{\mathcal{A}} \exp(\frac{1}{\alpha} Q_\phi(s, \bar{a})) d\bar{a}$ depends exclusively on the fixed critic parameters ϕ and the state s , it functions as a constant relative to the policy optimization parameters θ . Consequently, factoring out the linear scalar $\frac{1}{\alpha}$ allows us to rewrite the full expectation over the dataset $\mathcal{D}$ line-by-line:

$$\begin{aligned} \mathbb{E}_{s \sim \mathcal{D}} \left[D_{KL} \left(\pi_\theta(\cdot|s) \parallel \frac{\exp(\frac{1}{\alpha} Q_\phi(s, \cdot))}{Z_\phi(s)} \right) \right] &= \mathbb{E}_{s \sim \mathcal{D}, a \sim \pi_\theta} \left[\log \pi_\theta(a|s) - \frac{1}{\alpha} Q_\phi(s, a) \right] + \mathbb{E}_{s \sim \mathcal{D}} [\log Z_\phi(s)] \\ &= \frac{1}{\alpha} \mathbb{E}_{s \sim \mathcal{D}, a \sim \pi_\theta} [\alpha \log \pi_\theta(a|s) - Q_\phi(s, a)] + \mathbb{E}_{s \sim \mathcal{D}} [\log Z_\phi(s)] \\ &= \frac{1}{\alpha} J_\pi(\theta) + \text{const.} \end{aligned}$$

Because $\alpha > 0$, minimizing the total expected KL divergence with respect to θ is mathematically identical to minimizing the primary Soft Actor-Critic loss $J_\pi(\theta)$. Intuitively, this update directly encapsulates the soft policy improvement step; since the ideal unconstrained soft policy improvement mandates setting the policy to the exact Boltzmann form, minimizing this KL divergence represents a parametric projection of that optimal update onto the feasible space of the neural network policy π_θ .

2.6 Reparameterized Gradient

Question 9: REINFORCE Variance

[2 Points]

Explain why REINFORCE has high variance. Your answer should compare score-function gradients with pathwise gradients and should mention why long-horizon returns make the variance problem worse.

Answer:

The high variance characteristic of the REINFORCE estimator stems fundamentally from its reliance on the score-function gradient framework. Intuitively, the score function updates parameters by scaling log probabilities with the total sample trajectory return $R(\tau) = \sum_{t=0}^{T-1} \gamma^t r_t$, evaluating actions purely based on whether the final realization was positive or negative. Furthermore, as the optimization horizon increases, the return compounding across long sequences aggregates the random fluctuations of environmental transitions and policy choices from numerous time steps, causing the estimator's variance to explode.

In contrast, pathwise gradients utilize the reparameterization trick ($a_t = f_\theta(\epsilon_t; s_t)$), which decouples the policy parameters from the stochastic action execution distribution. This structural change allows gradients to backpropagate directly through the critic network via the analytical directional derivative:

$$\nabla_\theta Q_\phi(s, a) = \nabla_a Q_\phi(s, a) \cdot \nabla_\theta f_\theta(\epsilon; s).$$

Consequently, instead of treating the reward surface as a black box via sample trajectories, the pathwise gradient leverages the smooth, learned action-value function Q_ϕ . Because the critic network acts as a conditional expectation smoother over future states, it effectively averages out the catastrophic multi-step rollout noise, yielding dramatically reduced variance compared to score-function approaches.

Question 10: Reparameterized Gradient

[5 Points]

The SAC actor objective can be written as

$$J_\pi(\theta) = \mathbb{E}_{s_t \sim \mathcal{D}, \epsilon_t \sim \mathcal{N}} [\alpha \log \pi_\theta(a_t | s_t) - Q_\phi(s_t, a_t)],$$

Derive the full reparameterized gradient

$$\nabla_\theta J_\pi(\theta) = \mathbb{E}_{s_t \sim \mathcal{D}, \epsilon_t \sim \mathcal{N}} [\nabla_\theta \alpha \log \pi_\theta(a_t | s_t) - \nabla_{a_t} Q_\phi(s_t, a_t) \nabla_\theta f_\theta].$$

Answer:

To derive the pathwise policy gradient, we consider the parameterization of the continuous action variable given by $a_t = f_\theta(\epsilon_t; s_t)$, where the stochasticity is governed by a parameter-independent baseline distribution $\epsilon_t \sim \mathcal{N}(0, I)$. Because the distributions generating the state history $s_t \sim \mathcal{D}$ and the noise vector $\epsilon_t \sim \mathcal{N}$ are decoupled from the parameter vector θ , the gradient operator ∇_θ can be moved directly inside the expectation terms via Leibniz's rule:

$$\nabla_\theta J_\pi(\theta) = \mathbb{E}_{s_t \sim \mathcal{D}, \epsilon_t \sim \mathcal{N}} [\nabla_\theta (\alpha \log \pi_\theta(a_t | s_t) - Q_\phi(s_t, a_t))].$$

We expand the total derivative with respect to θ for each component inside the integrand by applying the multivariate chain rule. For the first term, the policy parameters θ influence $\log \pi_\theta(a_t|s_t)$ both directly through its functional definition and implicitly through the parameterized action argument a_t :

$$\nabla_\theta (\alpha \log \pi_\theta(a_t|s_t)) = \nabla_\theta \alpha \log \pi_\theta(a|s_t) \Big|_{a=a_t} + \nabla_{a_t} \alpha \log \pi_\theta(a_t|s_t) \cdot \nabla_\theta f_\theta(\epsilon_t; s_t).$$

For the second term, since the critic network parameterization ϕ is held fixed during the actor update phase, the parameter vector θ only influences the action-value estimate through the input action argument a_t :

$$\nabla_\theta Q_\phi(s_t, a_t) = \nabla_{a_t} Q_\phi(s_t, a_t) \cdot \nabla_\theta f_\theta(\epsilon_t; s_t).$$

Furthermore, by substituting these individual chain rule expansions back into the primary expectation equation and grouping the implicit structural derivatives that flow through the reparameterized action function, the full gradient expression is recovered:

$$\nabla_\theta J_\pi(\theta) = \mathbb{E}_{s_t \sim \mathcal{D}, \epsilon_t \sim N} [\nabla_\theta \alpha \log \pi_\theta(a_t|s_t) + (\nabla_{a_t} \alpha \log \pi_\theta(a_t|s_t) - \nabla_{a_t} Q_\phi(s_t, a_t)) \nabla_\theta f_\theta(\epsilon_t; s_t)].$$

Intuitively, standard implementations of Soft Actor-Critic omit the explicit evaluation of the $\nabla_{a_t} \alpha \log \pi_\theta(a_t|s_t)$ component when calculating the inner pathwise action derivative, matching the canonical target formulation given by:

$$\nabla_\theta J_\pi(\theta) = \mathbb{E}_{s_t \sim \mathcal{D}, \epsilon_t \sim N} [\nabla_\theta \alpha \log \pi_\theta(a_t|s_t) - \nabla_{a_t} Q_\phi(s_t, a_t) \nabla_\theta f_\theta],$$

which completes the exact analytical derivation.

Question 11: Squashed Gaussian Policy and Change of Variables

[3 Points]

In SAC, continuous actions are often sampled using a squashed Gaussian policy:

$$u_\theta(s, \epsilon) = \mu_\theta(s) + \sigma_\theta(s) \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

$$a = \tanh(u_\theta(s, \epsilon)).$$

Derive the log-probability correction

$$\log \pi_\theta(a|s) = \log \mathcal{N}(u; \mu_\theta(s), \sigma_\theta(s)^2) - \sum_{j=1}^d \log(1 - \tanh^2(u_j)),$$

where d is the action dimension.

Answer:

To derive the log-probability density of the squashed continuous action, we utilize the multi-dimensional change of variables theorem. Given that the transformation $a = \tanh(u)$ is an invertible, element-wise bijective mapping between the unbounded latent space and the bounded action space, the probability density $\pi_\theta(a|s)$ is structurally linked to the underlying Gaussian den-

sity $p(u) = \mathcal{N}(u; \mu_\theta(s), \sigma_\theta(s)^2)$ via the absolute determinant of the Jacobian matrix:

$$\pi_\theta(a|s) = p(u) \cdot \left| \det \left(\frac{da}{du} \right) \right|^{-1}.$$

Since the squashing function is applied independently to each individual component of the vector ($a_j = \tanh(u_j)$ for $j = 1, \dots, d$), the resulting Jacobian matrix $\frac{da}{du}$ is strictly diagonal. Consequently, its determinant simplifies directly to the product of its diagonal scalar derivatives:

$$\det \left(\frac{da}{du} \right) = \prod_{j=1}^d \frac{da_j}{du_j} = \prod_{j=1}^d (1 - \tanh^2(u_j)).$$

Substituting this diagonal determinant expression back into the density transformation formula yields:

$$\pi_\theta(a|s) = p(u) \cdot \prod_{j=1}^d \frac{1}{1 - \tanh^2(u_j)}.$$

Furthermore, taking the natural logarithm of both sides converts the continuous product operation into a discrete summation. The algebraic expansion simplifies line-by-line as follows:

$$\begin{aligned} \log \pi_\theta(a|s) &= \log p(u) + \log \left(\prod_{j=1}^d \frac{1}{1 - \tanh^2(u_j)} \right) \\ &= \log \mathcal{N}(u; \mu_\theta(s), \sigma_\theta(s)^2) + \sum_{j=1}^d \log \left(\frac{1}{1 - \tanh^2(u_j)} \right) \\ &= \log \mathcal{N}(u; \mu_\theta(s), \sigma_\theta(s)^2) - \sum_{j=1}^d \log(1 - \tanh^2(u_j)). \end{aligned}$$

Intuitively, the subtraction of the log-determinant penalizes actions near the boundaries of the squashed space, accurately accounting for the volume compression caused by the tanh bounding function.

2.7 Temperature Tuning

Question 12: Automatic Temperature Tuning

[3 Points]

Consider the entropy constraint

$$\mathbb{E}_{s \sim \mathcal{D}, a \sim \pi_\theta(\cdot|s)} [-\log \pi_\theta(a|s)] \geq \bar{\mathcal{H}}.$$

Treating the policy π_θ as fixed for the temperature update, derive the temperature loss

$$J(\alpha) = \mathbb{E}_{s \sim \mathcal{D}, a \sim \pi_\theta(\cdot|s)} [-\alpha(\log \pi_\theta(a|s) + \bar{\mathcal{H}})], \quad \alpha > 0,$$

which is minimized with respect to α . Compute $\nabla_\alpha J(\alpha)$ and explain how gradient descent on $J(\alpha)$ changes α when the policy entropy is below or above the target entropy.

Answer:

To derive the parametric gradient of the temperature objective $J(\alpha)$ with respect to α , we differentiate the loss function while treating the underlying policy π_θ as a fixed distribution. Since the expectation operator is evaluated over a stationary data distribution, the derivative operator passes directly inside the expectation terms via Leibniz's rule:

$$\nabla_\alpha J(\alpha) = \nabla_\alpha \mathbb{E}_{s \sim \mathcal{D}, a \sim \pi_\theta} [-\alpha(\log \pi_\theta(a|s) + \bar{\mathcal{H}})] = \mathbb{E}_{s \sim \mathcal{D}, a \sim \pi_\theta} [-(\log \pi_\theta(a|s) + \bar{\mathcal{H}})].$$

Distributing the expectation over the inner linear components enables the isolation of the canonical policy entropy term, defined mathematically as $\mathcal{H}(\pi_\theta(\cdot|s)) = \mathbb{E}_{a \sim \pi_\theta} [-\log \pi_\theta(a|s)]$:

$$\begin{aligned} \nabla_\alpha J(\alpha) &= -\mathbb{E}_{s \sim \mathcal{D}, a \sim \pi_\theta} [\log \pi_\theta(a|s)] - \mathbb{E}_{s \sim \mathcal{D}, a \sim \pi_\theta} [\bar{\mathcal{H}}] \\ &= \mathbb{E}_{s \sim \mathcal{D}} [\mathbb{E}_{a \sim \pi_\theta} [-\log \pi_\theta(a|s)]] - \bar{\mathcal{H}} \\ &= \mathbb{E}_{s \sim \mathcal{D}} [\mathcal{H}(\pi_\theta(\cdot|s))] - \bar{\mathcal{H}}. \end{aligned}$$

Furthermore, we evaluate how a standard gradient descent update step, given by $\alpha \leftarrow \alpha - \eta \nabla_\alpha J(\alpha)$ where $\eta > 0$ is the learning rate, dynamically regulates the temperature parameter under alternating entropy constraints:

- ◇ **Policy Entropy Below Target** ($\mathbb{E}_s[\mathcal{H}] < \bar{\mathcal{H}}$): When the current expected policy entropy is lower than the required threshold, the derivative becomes strictly negative ($\nabla_\alpha J(\alpha) < 0$). Consequently, the gradient descent update subtracts a negative value, which increases α . Intuitively, a higher temperature coefficient places greater structural weight on the entropy bonus in the actor's objective, incentivizing the policy to become more stochastic and expand its exploration.
- ◇ **Policy Entropy Above Target** ($\mathbb{E}_s[\mathcal{H}] > \bar{\mathcal{H}}$): Conversely, when the expected policy entropy exceeds the target baseline, the derivative evaluates to a positive quantity ($\nabla_\alpha J(\alpha) > 0$). Consequently, the update rule subtracts from the current temperature value, causing α to decrease. Intuitively, reducing the temperature parameter minimizes the relative importance of the entropy regularizer, allowing the policy to focus predominantly on pure reward maximization.

3 Asynchronous Advantage Actor-Critic (A3C)

3.1 Variance Reduction in Policy Gradients

Question 1: Baselines, Bias, and the Optimal Baseline

[7 Points]

For a fixed state s , define the score vector

$$g_\theta(s, a) = \nabla_\theta \log \pi_\theta(a|s).$$

Part (a) Prove that subtracting any state-dependent baseline $b(s)$ does not introduce bias into the policy-gradient estimator. Specifically, show that

$$\mathbb{E}_{a \sim \pi_\theta(\cdot|s)} [g_\theta(s, a)b(s)] = 0.$$

Part (b) Now consider the single-state gradient estimator

$$X_b = g_\theta(s, a)(Q^{\pi_\theta}(s, a) - b(s)).$$

Derive the scalar baseline $b^*(s)$ that minimizes

$$\mathbb{E}_{a \sim \pi_\theta(\cdot|s)}[\|X_b\|_2^2].$$

Part (c) Is the optimal baseline $b^*(s)$ always equal to $V^{\pi_\theta}(s)$? If not, state a sufficient condition under which they are equal.

Answer:

Part (a) To demonstrate that the addition of a state-dependent baseline $b(s)$ maintains an unbiased policy gradient estimator, we evaluate the expected value of the baseline-weighted score function over the action distribution at state s . Expanding this expectation into its continuous integral form yields:

$$\mathbb{E}_{a \sim \pi_\theta(\cdot|s)}[g_\theta(s, a)b(s)] = \int_{\mathcal{A}} \pi_\theta(a|s) \nabla_\theta \log \pi_\theta(a|s) b(s) da.$$

By substituting the classical log-derivative identity $\nabla_\theta \log \pi_\theta(a|s) = \frac{\nabla_\theta \pi_\theta(a|s)}{\pi_\theta(a|s)}$, the explicit density terms cancel out. Furthermore, because the baseline $b(s)$ is strictly independent of the action variable a , it factors completely outside the integration operator:

$$\begin{aligned} \mathbb{E}_{a \sim \pi_\theta(\cdot|s)}[g_\theta(s, a)b(s)] &= \int_{\mathcal{A}} \pi_\theta(a|s) \frac{\nabla_\theta \pi_\theta(a|s)}{\pi_\theta(a|s)} b(s) da \\ &= b(s) \int_{\mathcal{A}} \nabla_\theta \pi_\theta(a|s) da \\ &= b(s) \nabla_\theta \int_{\mathcal{A}} \pi_\theta(a|s) da. \end{aligned}$$

Axiomatically, the total integral of a valid policy distribution over its complete action space must equal unity ($\int_{\mathcal{A}} \pi_\theta(a|s) da = 1$). Since the partial derivative of a constant value is identically zero, the expression simplifies directly:

$$b(s) \nabla_\theta(1) = 0.$$

Consequently, subtracting $b(s)$ from the action-value function yields the modified policy gradient component $\nabla_\theta J_{\text{new}}(\theta) = \nabla_\theta J_{\text{old}}(\theta) - \mathbb{E}_s[b(s) \mathbb{E}_a[g_\theta(s, a)]] = \nabla_\theta J_{\text{old}}(\theta)$, proving that the baseline introduces zero analytical bias into the gradient field.

Part (b) To derive the optimal variance-minimizing scalar baseline $b^*(s)$, we minimize the expected squared L_2 -norm of the single-state gradient estimator X_b . Expanding this objective function via continuous integration gives:

$$\mathbb{E}_{a \sim \pi_\theta(\cdot|s)}[\|X_b\|_2^2] = \int_{\mathcal{A}} \pi_\theta(a|s) \|g_\theta(s, a)\|_2^2 (Q^{\pi_\theta}(s, a) - b(s))^2 da.$$

Taking the formal derivative of this objective with respect to the scalar baseline parameter $b(s)$ and setting it to zero defines the necessary first-order optimality condition:

$$\frac{d}{db(s)} \mathbb{E}_{a \sim \pi_\theta(\cdot|s)}[\|X_b\|_2^2] = \int_{\mathcal{A}} \pi_\theta(a|s) \|g_\theta(s, a)\|_2^2 \cdot 2(b(s) - Q^{\pi_\theta}(s, a)) da = 0.$$

Distributing the integration operator over the inner linear components allows us to separate the terms:

$$2b(s) \int_{\mathcal{A}} \pi_\theta(a|s) \|g_\theta(s, a)\|_2^2 da - 2 \int_{\mathcal{A}} \pi_\theta(a|s) Q^{\pi_\theta}(s, a) \|g_\theta(s, a)\|_2^2 da = 0.$$

Converting these individual integrals back into their canonical expected value notations yields the following balance equation:

$$b(s)\mathbb{E}_{a\sim\pi_\theta(\cdot|s)}[\|g_\theta(s,a)\|_2^2] = \mathbb{E}_{a\sim\pi_\theta(\cdot|s)}[Q^{\pi_\theta}(s,a)\|g_\theta(s,a)\|_2^2].$$

Moreover, isolating the scalar baseline term completes the optimization, providing the definitive variance-minimizing formulation:

$$b^*(s) = \frac{\mathbb{E}_{a\sim\pi_\theta(\cdot|s)}[Q^{\pi_\theta}(s,a)\|g_\theta(s,a)\|_2^2]}{\mathbb{E}_{a\sim\pi_\theta(\cdot|s)}[\|g_\theta(s,a)\|_2^2]}.$$

Part (c) Intuitively, the optimal baseline $b^*(s)$ is not universally equivalent to the standard state-value function $V^{\pi_\theta}(s) = \mathbb{E}_{a\sim\pi_\theta}[Q^{\pi_\theta}(s,a)]$. Mechanically, $b^*(s)$ represents an expected action-value weighted by the squared magnitude of the score function, prioritizing regions where the policy parameters exhibit high directional sensitivity.

Furthermore, a sufficient condition for their equivalence arises when the squared L_2 -norm of the score vector is entirely invariant with respect to the action argument a for a given state s . Mathematically, this condition is satisfied if $\|g_\theta(s,a)\|_2^2 = C(s)$ for some action-independent scalar function $C(s)$. Substituting this invariant function into the optimal baseline expression allows the scalar to factor out of the expectations:

$$b^*(s) = \frac{\mathbb{E}_{a\sim\pi_\theta(\cdot|s)}[Q^{\pi_\theta}(s,a) \cdot C(s)]}{\mathbb{E}_{a\sim\pi_\theta(\cdot|s)}[C(s)]} = \frac{C(s)\mathbb{E}_{a\sim\pi_\theta(\cdot|s)}[Q^{\pi_\theta}(s,a)]}{C(s)\mathbb{E}_{a\sim\pi_\theta(\cdot|s)}[1]} = \mathbb{E}_{a\sim\pi_\theta(\cdot|s)}[Q^{\pi_\theta}(s,a)] = V^{\pi_\theta}(s).$$

Consequently, the variance-minimizing baseline collapses identically to the true soft state-value function if and only if the score function magnitude is uniform across the local action space.

3.2 Multi-Step Advantage Estimation

Question 2: Multi-Step Advantage as a Sum of TD Errors

[3 Points]

Show that the n-step advantage estimator can be written as

$$\hat{A}_t^{(n)} = R_t^{(n)} - V_\phi(s_t) = \sum_{k=0}^{n-1} \gamma^k \delta_{t+k}.$$

Then explain how increasing n affects the bias and variance of the advantage estimator.

Answer:

We expand the discounted sum of one-step temporal-difference errors by substituting the definition $\delta_{t+k} = r_{t+k} + \gamma V_\phi(s_{t+k+1}) - V_\phi(s_{t+k})$:

$$\sum_{k=0}^{n-1} \gamma^k \delta_{t+k} = \sum_{k=0}^{n-1} \gamma^k r_{t+k} + \sum_{k=0}^{n-1} \gamma^{k+1} V_\phi(s_{t+k+1}) - \sum_{k=0}^{n-1} \gamma^k V_\phi(s_{t+k}).$$

Since the value function terms constitute a classic telescoping series, the summation reduces strictly to its boundary values, yielding:

$$\sum_{k=0}^{n-1} \gamma^{k+1} V_{\phi}(s_{t+k+1}) - \sum_{k=0}^{n-1} \gamma^k V_{\phi}(s_{t+k}) = \gamma^n V_{\phi}(s_{t+n}) - V_{\phi}(s_t).$$

Substituting this boundary collapse back and employing the canonical definition of the n -step return target, $R_t^{(n)} = \sum_{k=0}^{n-1} \gamma^k r_{t+k} + \gamma^n V_{\phi}(s_{t+n})$, directly recovers the advantage formulation:

$$\sum_{k=0}^{n-1} \gamma^k \delta_{t+k} = \sum_{k=0}^{n-1} \gamma^k r_{t+k} + \gamma^n V_{\phi}(s_{t+n}) - V_{\phi}(s_t) = R_t^{(n)} - V_{\phi}(s_t) = \hat{A}_t^{(n)}.$$

Furthermore, increasing the temporal horizon n modulates the estimator's statistical profile. Intuitively, a larger n reduces estimation bias because the target relies more heavily on actual realized rewards rather than the bootstrapped critic, decoupling updates from local function approximation errors. Consequently, this extended window simultaneously scales up variance, as longer multi-step rollouts aggregate a larger sequence of stochastic environmental transitions and cumulative sample trajectory noise.

Question 3: Critic-Induced Bias in Multi-Step Actor-Critic

[5 Points]

Assume that the learned critic differs from the true value function by an approximation error

$$V_{\phi}(s) = V^{\pi}(s) + e(s).$$

For $n \geq 1$ derive an explicit expression for the bias

$$\mathbb{E}[\hat{A}_t^{(n)} | s_t = s, a_t = a] - A^{\pi}(s, a).$$

Your final answer should be written in terms of $e(s)$, the one-step transition kernel $P(\cdot | s, a)$, and the policy-induced transition matrix P_{π} . Then answer the following:

1. What does the bias become when $n = 1$?
2. What happens to the bias as $n \rightarrow \infty$, assuming $\gamma^n e(s_{t+n}) \rightarrow 0$?
3. Why does the term $-e(s)$ alone not bias the policy gradient?
4. Why can the next-state error term bias the actor?

Answer:

The n -step advantage estimator is defined as $\hat{A}_t^{(n)} = R_t^{(n)} - V_{\phi}(s_t)$, where $R_t^{(n)} = \sum_{k=0}^{n-1} \gamma^k r_{t+k} + \gamma^n V_{\phi}(s_{t+n})$. To derive the explicit expression for the critic-induced bias, we evaluate the expected value of $\hat{A}_t^{(n)}$ conditioned on the current state-action pair ($s_t = s, a_t = a$):

$$\mathbb{E}[\hat{A}_t^{(n)} | s_t = s, a_t = a] = \mathbb{E}\left[\sum_{k=0}^{n-1} \gamma^k r_{t+k} \mid s_t = s, a_t = a\right] + \gamma^n \mathbb{E}[V_{\phi}(s_{t+n}) | s_t = s, a_t = a] - \mathbb{E}[V_{\phi}(s_t) | s_t = s].$$

Recognizing that the expected reward rollout satisfies $\mathbb{E}\left[\sum_{k=0}^{n-1} \gamma^k r_{t+k} \mid s_t = s, a_t = a\right] = Q^{\pi}(s, a) - \gamma^n \mathbb{E}[V^{\pi}(s_{t+n}) | s_t = s, a_t = a]$, we substitute the function approximation error model $V_{\phi}(s) =$

$V^\pi(s) + e(s)$ directly into the target expression:

$$\begin{aligned}\mathbb{E} \left[\hat{A}_t^{(n)} \mid s_t = s, a_t = a \right] &= Q^\pi(s, a) - \gamma^n \mathbb{E} [V^\pi(s_{t+n}) \mid s_t = s, a_t = a] \\ &\quad + \gamma^n \mathbb{E} [V^\pi(s_{t+n}) + e(s_{t+n}) \mid s_t = s, a_t = a] - (V^\pi(s) + e(s)).\end{aligned}$$

Simplifying this expression line-by-line causes the true multi-step state-value expectations to cancel out identically:

$$\begin{aligned}&= Q^\pi(s, a) - V^\pi(s) + \gamma^n \mathbb{E} [e(s_{t+n}) \mid s_t = s, a_t = a] - e(s) \\ &= A^\pi(s, a) + \gamma^n \mathbb{E} [e(s_{t+n}) \mid s_t = s, a_t = a] - e(s).\end{aligned}$$

Consequently, subtracting the true advantage $A^\pi(s, a)$ yields the explicit advantage bias expression:

$$\text{Bias}^{(n)}(s, a) = \gamma^n \mathbb{E} [e(s_{t+n}) \mid s_t = s, a_t = a] - e(s).$$

To formulate this expectation using state transition operators, the first time step transitions from state s via action a through the one-step environment kernel $P(\cdot|s, a)$. The remaining $n-1$ temporal steps follow the policy-induced transition matrix P_π . Thus, expanding the expectation into integral form yields:

$$\text{Bias}^{(n)}(s, a) = \gamma^n \int_{\mathcal{S}} P(s'|s, a) \int_{\mathcal{S}} P_\pi^{n-1}(s''|s') e(s'') ds'' ds' - e(s).$$

1. **Bias for $n = 1$:** When $n = 1$, the multi-step policy transition operator reduces to the identity matrix ($P_\pi^0 = \mathbf{I}$). Consequently, the explicit bias expression simplifies directly to:

$$\text{Bias}^{(1)}(s, a) = \gamma \int_{\mathcal{S}} P(s'|s, a) e(s') ds' - e(s).$$

2. **Asymptotic Limit ($n \rightarrow \infty$):** As $n \rightarrow \infty$, under the axiomatic damping assumption that $\gamma^n e(s_{t+n}) \rightarrow 0$, the first-term multi-step expectation collapses to zero. Consequently, the asymptotic bias isolates entirely to the current state error:

$$\lim_{n \rightarrow \infty} \text{Bias}^{(n)}(s, a) = -e(s).$$

3. **Unbiased Nature of $-e(s)$:** The standalone current-state error term $-e(s)$ does not introduce bias into the computed policy gradient because it is completely action-independent. Intuitively, when evaluating the expected value of the score function, it acts as a state-dependent constant baseline and factors outside the action integral, evaluating identically to zero:

$$\mathbb{E}_{a \sim \pi_\theta(\cdot|s)} [\nabla_\theta \log \pi_\theta(a|s) \cdot (-e(s))] = -e(s) \nabla_\theta \int_{\mathcal{A}} \pi_\theta(a|s) da = -e(s) \nabla_\theta(1) = 0.$$

4. **Biased Nature of Next-State Error:** Conversely, the next-state error term biases the actor because the state transition probability distribution $P(s'|s, a)$ is directly dependent on the executed action a . Consequently, it cannot be factored out of the policy expectation operator, structurally altering the gradient coordinates and leading to a distorted, suboptimal policy update direction.

3.3 Synchronous Advantage Actor-Critic (A2C)

The Synchronous Advantage Actor-Critic (A2C) algorithm formalizes a parallel optimization framework where parametric actor and critic networks are updated synchronously using multi-step bootstrapped targets. By coordinating parameter updates across discrete environmental rollouts, A2C avoids the complex coordination overhead of asynchronous updates. Consequently, it maintains mathematically stable gradient trajectories while optimizing both state-value predictions and policy tracking configurations.

Question 4: A2C Update Equations and Semi-Gradient Critic Learning [3 Points]

Part (a) Derive the semi-gradient update for the critic: $\nabla_{\phi} L_{\text{critic}}(\phi)$. In this derivation, treat the bootstrap target $\hat{R}_t$ as a constant with respect to ϕ . Part (b) Derive the policy-gradient update for the actor: $\nabla_{\theta} J_{\text{actor}}(\theta)$. In this derivation, treat the advantage estimate $\hat{A}_t^{(n)}$ as a constant with respect to θ . Part (c) Explain why the critic update above is called a semi-gradient update. What term would appear if we fully differentiated through the bootstrap target?

Answer:

Part (a) We evaluate the derivative of the mean-squared value error loss function $L_{\text{critic}}(\phi) = \mathbb{E} \left[\frac{1}{2} (V_{\phi}(s_t) - \hat{R}_t)^2 \right]$ with respect to the critic parameters ϕ . Moving the gradient operator inside the expectation via Leibniz's rule and applying the standard power rule yields:

$$\nabla_{\phi} L_{\text{critic}}(\phi) = \mathbb{E} \left[\nabla_{\phi} \frac{1}{2} (V_{\phi}(s_t) - \hat{R}_t)^2 \right] = \mathbb{E} \left[(V_{\phi}(s_t) - \hat{R}_t) \nabla_{\phi} (V_{\phi}(s_t) - \hat{R}_t) \right].$$

Treating the target $\hat{R}_t$ as an independent parameter-free scalar constant implies $\nabla_{\phi} \hat{R}_t = 0$. Thus, the final analytical semi-gradient expression reduces line-by-line to:

$$\nabla_{\phi} L_{\text{critic}}(\phi) = \mathbb{E} \left[(V_{\phi}(s_t) - \hat{R}_t) \nabla_{\phi} V_{\phi}(s_t) \right].$$

Part (b) To obtain the actor network update direction, we evaluate the parametric gradient of the advantage-weighted objective function $J_{\text{actor}}(\theta) = \mathbb{E} \left[\log \pi_{\theta}(a_t | s_t) \hat{A}_t^{(n)} \right]$. Treating the computed multi-step advantage estimate $\hat{A}_t^{(n)}$ as a fixed conditional modifier relative to θ allows it to pass through the parameter gradient operator as a scalar baseline coefficient:

$$\nabla_{\theta} J_{\text{actor}}(\theta) = \mathbb{E} \left[\nabla_{\theta} \left(\log \pi_{\theta}(a_t | s_t) \hat{A}_t^{(n)} \right) \right] = \mathbb{E} \left[\nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \hat{A}_t^{(n)} \right].$$

Part (c) The critic parameter parameterization is designated as a semi-gradient update because it intentionally omits the implicit dependency of the bootstrapping target $\hat{R}_t = \sum_{k=0}^{n-1} \gamma^k r_{t+k} + \gamma^n V_{\phi}(s_{t+n})$ on the parameter vector ϕ . Furthermore, if we fully differentiated through this temporal-difference target structure, an additional implicit chain rule component would emerge from the boundary state representation:

$$\nabla_{\phi} \hat{R}_t = \nabla_{\phi} \left(\sum_{k=0}^{n-1} \gamma^k r_{t+k} + \gamma^n V_{\phi}(s_{t+n}) \right) = \gamma^n \nabla_{\phi} V_{\phi}(s_{t+n}).$$

Consequently, accounting for this downstream feedback propagation would introduce the following additional matrix term into the primary expected gradient calculation field:

$$\text{Extra Term} = -\gamma^n \mathbb{E} \left[(V_{\phi}(s_t) - \hat{R}_t) \nabla_{\phi} V_{\phi}(s_{t+n}) \right].$$

3.4 Entropy Regularization

Question 5: Entropy as KL Regularization

[2 Points]

Part (a) Show that maximizing entropy is equivalent to minimizing the KL divergence from the policy to the uniform distribution over actions: $D_{KL}(\pi_\theta(\cdot|s) \parallel \text{Unif}(\mathcal{A}))$. Part (b) For a discrete action space with A actions, derive $\nabla_\theta H(\pi_\theta(\cdot|s)) = -\sum_{a \in \mathcal{A}} \nabla_\theta \pi_\theta(a|s)(\log \pi_\theta(a|s) + 1)$. Part (c) Explain why entropy regularization is particularly important in A3C, where each worker collects on-policy data without an experience replay buffer.

Answer:

Part (a) Let $|\mathcal{A}|$ denote the total cardinality of a bounded discrete action space, yielding the constant probability mass function $\text{Unif}(a) = \frac{1}{|\mathcal{A}|}$. Expanding the definition of the Kullback-Leibler (KL) divergence from the policy distribution $\pi_\theta(\cdot|s)$ to this uniform reference distribution gives:

$$D_{KL}(\pi_\theta(\cdot|s) \parallel \text{Unif}(\mathcal{A})) = \sum_{a \in \mathcal{A}} \pi_\theta(a|s) \log \left(\frac{\pi_\theta(a|s)}{1/|\mathcal{A}|} \right) = \sum_{a \in \mathcal{A}} \pi_\theta(a|s) [\log \pi_\theta(a|s) + \log |\mathcal{A}|].$$

Distributing the linear summation across both inner mathematical terms and invoking the standard normalization axiom $\sum_{a \in \mathcal{A}} \pi_\theta(a|s) = 1$ yields:

$$= \sum_{a \in \mathcal{A}} \pi_\theta(a|s) \log \pi_\theta(a|s) + \log |\mathcal{A}| \sum_{a \in \mathcal{A}} \pi_\theta(a|s) = -H(\pi_\theta(\cdot|s)) + \log |\mathcal{A}|.$$

Since $\log |\mathcal{A}|$ represents an invariant, parameter-free constant relative to θ , minimizing the expected directional KL divergence is analytically identical to maximizing the structural policy entropy, establishing that $\arg \max_\theta H(\pi_\theta) \equiv \arg \min_\theta D_{KL}(\pi_\theta \parallel \text{Unif}(\mathcal{A}))$.

Part (b) We apply the multivariate product differentiation rule to the discrete entropy formulation $H(\pi_\theta(\cdot|s)) = -\sum_{a \in \mathcal{A}} \pi_\theta(a|s) \log \pi_\theta(a|s)$. Distributing the gradient operator ∇_θ element-wise across the individual components yields:

$$\nabla_\theta H(\pi_\theta(\cdot|s)) = -\sum_{a \in \mathcal{A}} \nabla_\theta [\pi_\theta(a|s) \log \pi_\theta(a|s)] = -\sum_{a \in \mathcal{A}} [\nabla_\theta \pi_\theta(a|s) \cdot \log \pi_\theta(a|s) + \pi_\theta(a|s) \cdot \nabla_\theta \log \pi_\theta(a|s)].$$

Substituting the classical score identity $\nabla_\theta \log \pi_\theta(a|s) = \frac{\nabla_\theta \pi_\theta(a|s)}{\pi_\theta(a|s)}$ cancels out the explicit denominator variables:

$$= -\sum_{a \in \mathcal{A}} \left[\nabla_\theta \pi_\theta(a|s) \log \pi_\theta(a|s) + \pi_\theta(a|s) \frac{\nabla_\theta \pi_\theta(a|s)}{\pi_\theta(a|s)} \right] = -\sum_{a \in \mathcal{A}} [\nabla_\theta \pi_\theta(a|s) \log \pi_\theta(a|s) + \nabla_\theta \pi_\theta(a|s)].$$

Factoring out the shared directional vector component $\nabla_\theta \pi_\theta(a|s)$ recovers the target closed-form expression:

$$\nabla_\theta H(\pi_\theta(\cdot|s)) = -\sum_{a \in \mathcal{A}} \nabla_\theta \pi_\theta(a|s)(\log \pi_\theta(a|s) + 1).$$

Part (c) Intuitively, the absence of an off-policy experience replay buffer in the A3C framework exposes individual learning workers to highly correlated, sequential on-policy data streams, which frequently induces severe optimization instability. Furthermore, without structural regularizers, the actor parameters rapidly succumb to positive feedback loops driven by localized rewards, causing the action selection distribution to prematurely collapse into sub-optimal deterministic modes.

Consequently, the entropy bonus acts as a continuous exploration safeguard; by penalizing sharp, low-entropy structures, it explicitly preserves random variance within individual rollout threads. This mechanism ensures that parallel workers actively explore distinct regions of the state space, preventing catastrophic global overfitting and stabilizing parameter trajectories.

3.5 Asynchronous Advantage Actor-Critic (A3C)

The Asynchronous Advantage Actor-Critic (A3C) framework utilizes multiple parallel worker threads interacting with localized environments to update a centralized set of global parameters. By decoupling data collection through asynchronous gradient optimization, A3C circumvents the requirement for an experience replay buffer. Consequently, this multi-worker configuration accelerates wall-clock execution time while structurally stabilizing on-policy learning trajectories. [cite: 556-562]

Question 6: Stale Gradients in Asynchronous Actor-Critic

[3 Points]

In A3C, a worker may compute a gradient using stale local parameters $\theta_{t-\tau}$ and apply it to the current global parameter θ_t . Let $G(\theta) = \nabla_{\theta} J(\theta)$ denote the exact policy-gradient field. Assume G is L -Lipschitz: $\|G(\theta) - G(\theta')\|_2 \leq L\|\theta - \theta'\|_2$. Assume each global parameter update satisfies $\|\theta_{k+1} - \theta_{k\text{Resize}}\|_2 \leq \alpha C$. Prove that the stale-gradient error is bounded by $\|G(\theta_t) - G(\theta_{t-\tau})\|_2 \leq L\alpha C\tau$. Then explain what this bound implies about the effect of the learning rate α , the delay τ , and the number of parallel workers.

Answer:

To derive the analytical bound on the stale-gradient error under a Lipschitz continuous gradient field, we express the total parameter divergence as a telescoping summation over the execution delay steps τ . Applying the generalized triangle inequality to this formulation yields:

$$\|G(\theta_t) - G(\theta_{t-\tau})\|_2 \leq \sum_{i=1}^{\tau} \|G(\theta_{t-i+1}) - G(\theta_{t-i})\|_2.$$

By invoking the L -Lipschitz continuity condition of the exact policy-gradient field $G(\theta)$, the point-wise distance between adjacent gradient configurations is bounded uniformly by the underlying parameter displacement:

$$\sum_{i=1}^{\tau} \|G(\theta_{t-i+1}) - G(\theta_{t-i})\|_2 \leq L \sum_{i=1}^{\tau} \|\theta_{t-i+1} - \theta_{t-i}\|_2.$$

Furthermore, substituting the bounded global parameter step condition $\|\theta_{k+1} - \theta_k\|_2 \leq \alpha C$ into the inductive summation simplifies the expression directly:

$$L \sum_{i=1}^{\tau} \|\theta_{t-i+1} - \theta_{t-i}\|_2 \leq L \sum_{i=1}^{\tau} \alpha C = L\alpha C\tau,$$

which completes the formal verification, establishing that $\|G(\theta_t) - G(\theta_{t-\tau})\|_2 \leq L\alpha C\tau$. [cite: 583-584]

Intuitively, this analytical upper bound reveals critical trade-offs regarding the structural hyper-parameters of the asynchronous optimization framework:

- ◊ **Learning Rate (α) and Per-step Bound (C):** Scaling down the learning rate α directly contracts the error bound, preventing destabilizing gradient updates at the cost of slower global parameter convergence.

- ◇ **Asynchronous Delay (τ):** The tracking error scales linearly with the update delay τ . Consequently, updates must be pushed rapidly to minimize parameter drift between local worker calculations and the central parameter server.
- ◇ **Parallel Workers (N):** Increasing the number of active parallel threads typically amplifies the gradient staleness parameter τ , since more workers overwrite global parameters simultaneously. Intuitively, this presents an intrinsic optimization trade-off between parallel data throughput and gradient correctness.

Question 7: Decorrelation Through Parallel Workers

[5 Points]

Suppose each A3C worker produces a stochastic gradient estimate g_i satisfying $\mathbb{E}[g_i] = g$, $\text{Var}(g_i) = \Sigma$. Assume that for $i \neq j$, $\text{Cov}(g_i, g_j) = \rho\Sigma$, where $\rho \in [0, 1]$ measures correlation between worker gradients. Let $\bar{g} = \frac{1}{N} \sum_{i=1}^N g_i$. Part (a) Compute $\text{Var}(\bar{g})$ in terms of N , ρ , and Σ . Part (b) Analyze the special cases $\rho = 0$ and $\rho = 1$. Part (c) Use this calculation to explain why running multiple parallel actors can stabilize on-policy learning without experience replay.

Answer:

Part (a) We evaluate the total variance of the empirical multi-worker average gradient $\bar{g}$ by distributing the linear variance operator across the individual stochastic components:

$$\text{Var}(\bar{g}) = \text{Var}\left(\frac{1}{N} \sum_{i=1}^N g_i\right) = \frac{1}{N^2} \text{Var}\left(\sum_{i=1}^N g_i\right).$$

Expanding the variance of a summation of correlated random variables decomposes the expression into individual variance fields and cross-covariance pairs:

$$= \frac{1}{N^2} \left[\sum_{i=1}^N \text{Var}(g_i) + \sum_{i \neq j} \text{Cov}(g_i, g_j) \right].$$

Given that there are exactly N individual variance terms and $N(N-1)$ unique cross-covariance terms, substituting the baseline matrices $\text{Var}(g_i) = \Sigma$ and $\text{Cov}(g_i, g_j) = \rho\Sigma$ yields:

$$\begin{aligned} \text{Var}(\bar{g}) &= \frac{1}{N^2} [N\Sigma + N(N-1)\rho\Sigma] \\ &= \frac{1}{N}\Sigma + \frac{N-1}{N}\rho\Sigma. \end{aligned}$$

Part (b) Evaluating the analytical boundary conditions for the gradient correlation parameter ρ reveals two distinct operational regimes:

- ◇ **Perfect Independence ($\rho = 0$):** Substituting $\rho = 0$ eliminates the downstream cross-covariance tracking term, reducing the overall variance expression line-by-line to:

$$\text{Var}(\bar{g}) = \frac{1}{N}\Sigma.$$

Consequently, under this regime, the total estimation variance decreases monotonically at a rate inversely proportional to the number of parallel workers N .

- ◇ **Perfect Correlation** ($\rho = 1$): Conversely, setting $\rho = 1$ forces the multi-worker variance expression to simplify via linear algebraic factoring:

$$\text{Var}(\bar{g}) = \frac{1}{N}\Sigma + \frac{N-1}{N}\Sigma = \Sigma.$$

Thus, when worker experiences are perfectly correlated, aggregating multiple parallel updates provides no statistical variance reduction, collapsing identically to a single localized rollout sequence.

Part (c) Intuitively, sequential on-policy data collection within a single centralized trajectory introduces intense temporal correlations, leading to high-variance gradient fields that destabilize training. Furthermore, while conventional setups deploy an experience replay buffer to homogenize data, replay mechanisms are inherently off-policy. [cite: 559-560]

Consequently, running multiple parallel actors interacting with independent environment instances provides an architectural method to minimize correlation ($\rho \rightarrow 0$). Because parallel threads traverse entirely distinct trajectories across the global state space, their stochastic gradient errors become statistically decoupled. As demonstrated in Part (b), when $\rho \approx 0$, the aggregate gradient variance diminishes scaled by $\frac{1}{N}$. This variance reduction smooths parameter updates, stabilizes the centralized on-policy learning trajectory, and enhances overall sample efficiency without requiring a memory-intensive off-policy experience replay structure. [cite: 597]